Hyperspectral Image Denoising Based on Global and Nonlocal Low-Rank Factorizations

Lina Zhuang, *Member, IEEE*, Xiyou Fu, Michael K. Ng, *Senior Member, IEEE*, and José M. Bioucas-Dias, *Fellow, IEEE*

***Abstract*— The ever-increasing spectral resolution of hyperspectral images (HSIs) is often obtained at the cost of a decrease in the signal-to-noise ratio of the measurements, thus calling for effective denoising techniques. HSIs from the real world lie in low-dimensional subspaces and are self-similar. The low dimensionality stems from the high correlation existing among the reflectance vectors, and self-similarity is common in real-world images. In this article, we exploit the above two properties. The low dimensionality is a global property that enables the denoising to be formulated just with respect to the subspace representation coefficients, thus greatly improving the denoising performance and reducing the computational complexity during processing. The self-similarity is exploited via a low-rank tensor factorization of nonlocal similar 3-D patches. The proposed factorization hinges on the optimal shrinkage/thresholding of the singular value decomposition (SVD) singular values of low-rank tensor unfoldings. As a result, the proposed method is user friendly and insensitive to its parameters. Its effectiveness is illustrated in a comparison with state-of-the-art competitors. A MATLAB demo of this work is available at https://github.com/LinaZhuang for the sake of reproducibility.**

***Index Terms*— 3-D patches, hyperspectral image (HSI) denoising, low-rank tensor factorization, self-similarity.**

I. Introduction

HYPERSPECTRAL cameras measure the electromagnetic energy scattered within their instantaneous field of view under hundreds or thousands of spectral channels with remarkably high spectral resolution, which enables precise material identification via spectroscopic analysis [1]. Hyperspectral remote sensing images have been widely used in countless applications (e.g., earth observation, environmental protection, and natural disaster monitoring). However, measurement noise often precludes the widespread use of hyperspectral images (HSIs) in precise material identification (e.g., precision farming) applications.

Hyperspectral denoising methods mainly exploit the high correlation of HSIs in the spatial and spectral domains. The high correlation allows extremely compact and sparse representations of HSIs on suitable frames where independent Gaussian noise is densely distributed. Furthermore, the images correspond to coefficients with large values on the frames, while Gaussian noise corresponds to near-zero coefficients. Therefore, we can perform image denoising by shrinking the coefficients of representations to attenuate the noise. This is the main idea of HSI denoising based on sparse representation.

HSIs can be sparsely represented in the gradient domain (total variation-based methods [2], [3]) on some preselected basis (such as a Fourier basis and Wavelet basis) [4] and on a data-adaptive basis (such as a low-dimensional subspace spanned by eigenvectors of data) [5], [6]. The sparser the representation of the HSI is, the more noise can be removed. Below, we introduce two characteristics of HSIs, namely, self-similarity in the spatial domain and low-rankness in the spatial–spectral domain. The characteristics have been exploited to create a compact and sparse representation of HSIs.

The self-similarity property of natural images has been exploited in state-of-the-art image denoising methods. The self-similarity of images means that they contain many similar patches at different locations or scales in the images. This characteristic has been recently exploited by patch-based image restoration methods, which are state-of-the-art in image denoising. Representative examples of this methodology in single-band images include the nonlocal means filter [7], the Gaussian mixture model (GMM) learned from a noisy image [8], the collaborative filtering of groups of similar patches block-matching and 3D filtering (BM3D) [9], and the weighted nuclear norm minimization (WNNM) [10]. Identical ideas have been pursued in multiband image denoising: block-matching and 4D filtering (BM4D) [11], Video-BM4D (VBM4D) [12], and multispectral principal component analysis-BM3D (MSPCA-BM3D) [13] use collaborative filtering in groups of 3-D patches extracted from volumetric data, videos, and multispectral data. higher order singular value decomposition (HOSVD) [14] applies a hard threshold filter to coefficients of HOSVD of similar patches. In HSI denoising, we refer to fast hyperspectral denoising (FastHyDe) [5] and RhyDe [6], which exploit the self-similarity of the HSI representation coefficients in suitable subspaces, non-local meets global (NG-meet) [15] that uses WNNM [10] to denoise similar image patches, and

Manuscript received August 20, 2020; revised November 9, 2020; accepted December 7, 2020. Date of publication January 8, 2021; date of current version November 24, 2021. This work was supported by the National Natural Science Foundation of China under Grant 42001287. The work of Michael K. Ng was supported in part by the Hong Kong Research Grants Council (HKRGC) General Research Fund (GRF) under Grant 12306616, Grant 12200317, Grant 12300218, Grant 12300519, and Grant 17201020 and in part by the University of Hong Kong (HKU) under Grant 104005583. *(Corresponding author: Xiyou Fu.)*

Lina Zhuang is with the Department of Mathematics, Hong Kong Baptist University, Kowloon, Hong Kong (e-mail: linazhuang@qq.com).

Xiyou Fu is with the Guangdong Laboratory of Artificial Intelligence and Digital Economy (SZ), Shenzhen University, Shenzhen 518060, China, and also with the College of Computer Science and Software Engineering, Shenzhen University, Shenzhen 518060, China (e-mail: fuxiyou@qq.com).

Michael K. Ng is with the Department of Mathematics, The University of Hong Kong, Pokfulam, Hong Kong (e-mail: mng@math.hku.hk).

José M. Bioucas-Dias is with the Instituto de Telecomunicações, Instituto Superior Técnico, Universidade de Lisbon, 1049-001 Lisbon, Portugal (e-mail: bioucas@lx.it.pt).

Digital Object Identifier 10.1109/TGRS.2020.3046038

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weighted low-rank tensor recovery (WLRTR) [16], which implements a WLRTR for similar image patches.

The high correlations among the spectral bands in local patches and groups of similar patches can be well characterized as low rankness in tensors/matrices:

1) The high-dimensional spectral vectors are represented on a low-dimensional subspace in a number of denoising methods [5], [6], [15], [17].
2) In the low-rank matrix recovery (LRMR) [18] and noise-adjusted iterative low-rank matrix approximation (NAILRMA) [19] methods, the local patches of an image are reshaped as column vectors and then stacked column by column to build a matrix (called Casorati matrix), which is a low-rank matrix because of the high correlation among the spectral bands in local patches.
3) A group of similar patches is modeled as a low-rank tensor in [16] and [17].

We can see that grouping similar patches makes better use of the correlation properties in HSIs than using only spectral vectors and local patches. However, a low-rank tensor approximation is still an active and challenging research topic with complexity issues, e.g., computing the rank of a tensor is NP-hard. Remarkable results have been achieved in low-rank matrix approximation. For example, works in [20], [21] and [22] analyze the effects of Gaussian noise on the SVD of a matrix using recent results from random matrix theory. However, the genuine extension of the matrix rank to tensors is challenging because many "obvious" properties in matrices cease to hold true [23] in tensors. In this work, instead of using the low-rank tensor approximation, we unfold the tensor into matrices along different modes successively and then apply the existing low-rank matrix technique.

The computational complexity of most HSI denoising methods is very high due to the large number of bands, usually on the order of hundreds. It happens, however, that hyperspectral vectors lie, with very good approximation, in subspaces of low dimension compared with the number of bands [1], and this subspace may be accurately inferred from the observed HSIs [24]. Subspace representations of HSIs have been employed to solve many inverse problems in hyperspectral imaging, such as image fusion [25], spectral unmixing [26], anomaly detection [27], hyperspectral segmentation [28], and classification [29]. Therefore, instead of denoising the original HSI, we can formulate the denoising with respect to the HSI subspace representation coefficients. This is the research line followed in [5], [6], [15], [30]and [2], and these methods are, to our knowledge, state of the art and faster than other methods.

The representation coefficient image (RCI) is formed by a few image bands, each linked with a subspace basis vector, hereafter termed *eigenimages*. FastHyDe denoises the eigenimages independently of each other, thus implicitly assuming that they are statistically independent. An informal justification for the independent processing is that the eigenimages are uncorrelated, owing to the use of eigenvectors of the signal correlation matrix as a subspace basis. However, uncorrelation does not imply independence. There is still room to improve the HSI denoising performance by taking the statistical dependence among the eigenimages into consideration.

Deep learning techniques also have been applied to hyperspectral noise reduction. Convolutional neural networks HSI-DeNet [31] and spatial-spectral gradient network (SSGN) [32] are two representative deep-denoisers showing impressive results. The deep-denoisers exploit spatial-spectral correlations in HSIs implicitly and their denoising performance highly depends on training data. A challenge of deep-denoisers is the lack of real HSIs that can be used as training data or how to generate simulated images close to real HSIs. But once a network is trained properly, deep-denoisers are much faster than the traditional machine learning (ML) based denoisers. Compared with deep-denoisers, ML-based denoisers do not need training data and usually can be applied to various HSIs robustly, but their high denoising accuracy usually comes with a high computational burden.

A. Contribution

This article exploits two main characteristics of hyperspectral data: 1) HSIs are well approximated by low-dimensional subspaces [1] and 2) HSIs are self-similar and thus eigenimages are also self-similar. The latter property is exploited via low-rank tensor factorization of nonlocal similar 3-D patches. The proposed factorization hinges on optimal shrinkage/thresholding of SVD singular values of the low-rank tensor unfoldings. As a result, the proposed method is user-friendly and insensitive to its parameters. This work is an extension of the material published in [33]. The new material includes the following.

1) Our method is introduced and described in more detail, and motivation for using SVD in three modes successively and some derivations of formulas are given.
2) An exhaustive array of experiments and comparisons is carried out. Eight representative or state-of-the-art HSI denoisers are compared with our method using two simulated HSIs and two real HSIs.

This article is organized as follows. Section II formulates the problem of HSI denoising. Section III introduces our new denoising algorithm step by step. Sections IV and V present the experimental results, including comparisons with the state-of-the-art methods. Section VI concludes this article.

II. Problem Formulation

A. Notations and Definitions

Vectors, matrices, and tensors are denoted by boldface lowercase letters (e.g., $\mathbf{x}$), boldface capital letters (e.g., $\mathbf{X}$), and calligraphic letters (e.g., $\mathcal{X}$), respectively.

Definition 1: The number of dimensions (ways) in a tensor is its ***order***, denoted by N. Each dimension (way) is called a ***mode***.

Definition 2: The mode-n vectors of $\mathcal{X}$ are defined as the I_n-dimensional vectors obtained from $\mathcal{X}$ by varying the index in i_n while keeping all the other indices fixed. For example, $\mathcal{X}(2, 3, :)$ is a mode-3 vector.

Definition 3: The i_nth mode-n slice of $\mathcal{X}$ is defined as an $(N-1)$th-order tensor obtained by fixing the mode-n index

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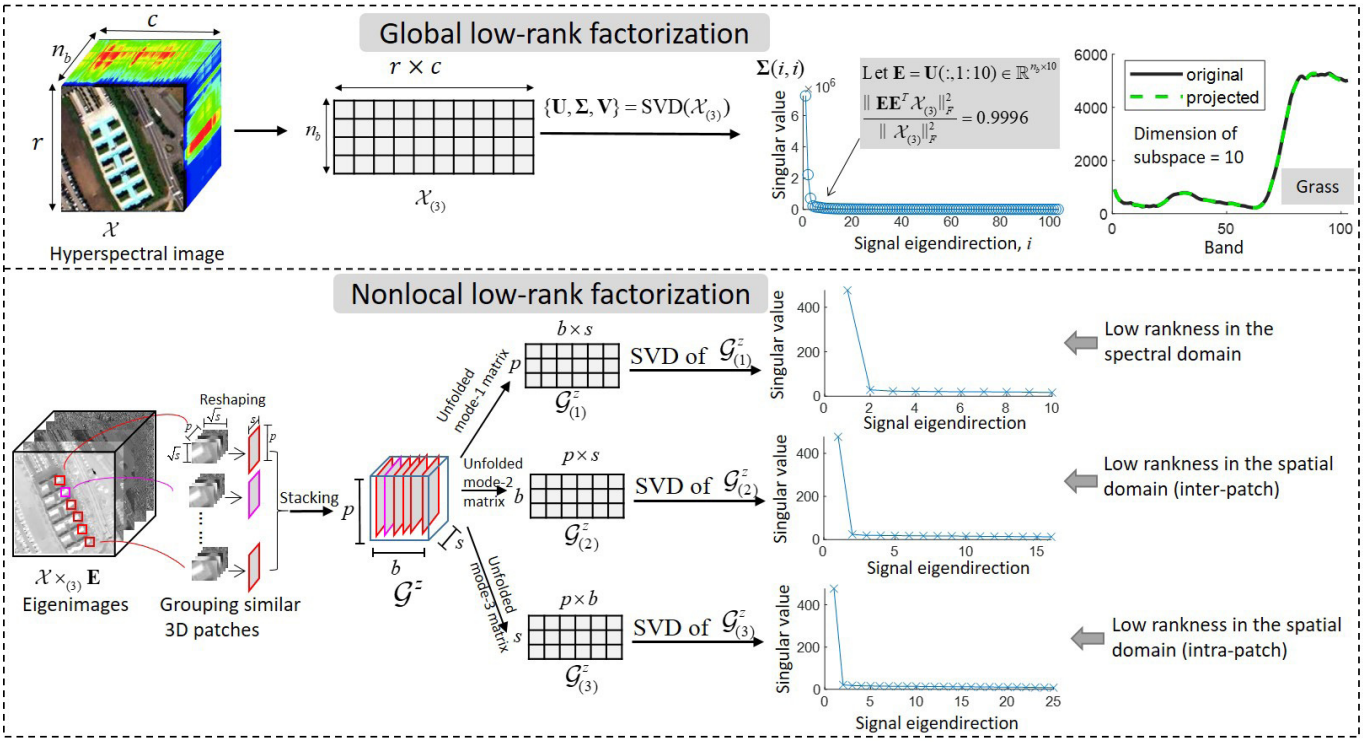


Fig. 1. Global low-rankness of a real HSI, Pavia University of size 614 × 340 × 103, and its nonlocal low-rankness. Only a small area of the image is displayed.

of $\mathcal{X}$ to be i_n: $\mathcal{X}(:,\ldots,:,i_n,:,\ldots,:)$. For example, $\mathcal{X}(:,:,1)$ is the first mode-3 slice.

Definition 4: (Unfolding: tensor to matrix transformation) A tensor can be unfolded into a matrix by rearranging its mode-n vectors. The mode-n unfolding of $\mathcal{X}$ is denoted by $\mathcal{X}_{(n)} \in \mathbb{R}^{I_n \times (I_1 \times \cdots \times I_{n-1} \times I_{n+1} \times \cdots \times I_N)}$, where the column vectors of $\mathcal{X}_{(n)}$ are the mode-n vectors of $\mathcal{X}$.

Definition 5: (Mode-n product: tensor matrix multiplication) The mode-n product of a tensor $\mathcal{X} \in \mathbb{R}^{I_1 \times I_2 \times \cdots \times I_N}$ by a matrix $\mathbf{E} \in \mathbb{R}^{J_n \times I_n}$ is a tensor $\mathcal{Y} \in \mathbb{R}^{I_1 \times \cdots \times I_{n-1} \times J_n \times I_{n+1} \times \cdots \times I_N}$, denoted as

$$\mathcal{Y} = \mathcal{X} \times_n \mathbf{E}$$

where each entry of $\mathcal{Y}$ is defined as the sum of the products of the corresponding entries in $\mathcal{X}$ and $\mathbf{E}$

$$\mathcal{Y}(i_1,\ldots,i_{n-1},j_n i_{n+1},\ldots,i_N) = \sum_{i_n} \mathcal{X}(i_1,\ldots,i_N) \cdot \mathbf{E}(j_n,i_n).$$

B. Observation Model

Let $\mathcal{X} \in \mathbb{R}^{r \times c \times n_b}$ denote a 3-D tensor representing an underlying clean HSI with $r \times c$ spectral vectors of size n_b (the number of bands in the sensor). Under the additive noise assumption, the observation model may be written as

$$\mathcal{Y} = \mathcal{X} + \mathcal{N} \tag{1}$$

where $\mathcal{Y}, \mathcal{N} \in \mathbb{R}^{r \times c \times n_b}$ represent the observed HSI data and noise, respectively. Entries in the noise tensor are assumed to be i.i.d. zero-mean Gaussian with variance σ^2. Let $\mathbf{C}_\lambda = \mathbb{E}[\mathbf{n}_i \mathbf{n}_i^{\mathrm{T}}]$ (where $\mathbf{n}_i$ is a generic column of unfolded mode-3 matrix $\mathcal{N}_{(3)}$) define the covariance matrix of the spectral noise. We have $\mathbf{C}_\lambda = \sigma^2 \mathbf{I}_{n_b}$ when the noise is i.i.d.

HSIs are high-dimensional data with a very high spectral resolution, leading to high correlations in the spectral domain. Spectral vectors (i.e., mode-3 vectors) of HSI $\mathcal{X}$ can be well represented in a low-dimensional subspace, which can be estimated from observed image $\mathcal{Y}$ with good approximation [1], [5], [6].

We take a real HSI, Pavia University data, as an example. This image is of high quality and considered a relatively clean image. We perform SVD of its unfolded mode-3 matrix, $\mathcal{X}_{(3)}$, whose columns are spectral vectors of pixels. We can see from Fig. 1 that the singular values decrease to near-zero rapidly. If we project the spectral vectors on a subspace, denoted by $\mathbf{E}$, spanned by ten singular vectors corresponding to the largest ten singular values, then we can compute the relative power of spectral vectors lying on the subspace: $\|\mathbf{E}\mathbf{E}^{\mathrm{T}}\mathcal{X}_{(3)}\|_F^2 / \|\mathcal{X}_{(3)}\|_F^2 = 0.9996$, implying that majority of information lies in the subspace. A spectral vector of grass in the image and its projection are plotted in Fig. 1, where we can infer that spectral vectors can be represented very well in an orthogonal subspace learned from the original HSI.

Based on the subspace representation of spectral vectors, we write $\mathcal{X} = \mathcal{Z} \times_3 \mathbf{E}$, with $\mathbf{E} \in \mathbb{R}^{n_b \times p}$ and $p \ll n_b$, and $\mathbf{E}$ holds an orthogonal basis for the signal subspace. Hence, the observation model (1) may be written as

$$\mathcal{Y} = \mathcal{Z} \times_3 \mathbf{E} + \mathcal{N}. \tag{2}$$

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Hereafter, $\mathcal{Z} \in \mathbb{R}^{r\times c\times p}$ is termed the RCI and its mode-3 slices are the eigenimages.

C. Eigenimage Denoising

Assuming that the noise is zero-mean Gaussian independent and identically distributed (other covariance matrices are easily dealt with [5]) and that $\mathbf{E}$ was learned from observated matrix $\mathcal{Y}_{(3)}$ [24] (see Section III-A for more details), the eigenimage denoising problem is formulated as

$$\widehat{\mathcal{Z}} = \arg\min_{\mathcal{Z}} \frac{1}{2}\|\mathcal{Y} - \mathcal{Z}\times_3\mathbf{E}\|_F^2 + \lambda\phi(\mathcal{Z}) \tag{3}$$

$$= \arg\min_{\mathcal{Z}} \frac{1}{2}\|\mathcal{Y}\times_3\mathbf{E}^{\mathrm{T}} - \mathcal{Z}\|_F^2 + \lambda\phi(\mathcal{Z}) \tag{4}$$

where $\|\mathcal{X}\|_F^2$ denotes the Frobenius norm that returns the square root of the sum of the absolute squares of its elements. The first term on the right-hand side of (3) represents the data fidelity and accounts for the zero-mean Gaussian i.i.d. noise. The second term in (3) is a regularizer expressing prior information tailored to self-similar eigenimages. The optimization in (3) is equivalent to that in (4) (see proof in Appendix A).

We remark that (4) is equivalent to a denoising problem where the additive noise is $\mathcal{N}\times_3\mathbf{E}^{\mathrm{T}}$; thus, the spectral covariance is $\mathbb{E}[\mathbf{E}\mathbf{n}_i(\mathbf{E}\mathbf{n}_i)^{\mathrm{T}}] = \mathbf{E}\mathbb{E}[\mathbf{n}_i\mathbf{n}_i^{\mathrm{T}}]\mathbf{E}^{\mathrm{T}} = \mathbf{E}(\sigma^2\mathbf{I}_{n_b})\mathbf{E}^{\mathrm{T}} = \sigma^2\mathbf{I}_p$, where $\mathbf{I}_p$ is the identity matrix of size specified in its subscript. Therefore, the noise attenuation resulting from the subspace representation is

$$\frac{\mathbb{E}\left[\|\mathcal{N}\times_3\mathbf{E}^{\mathrm{T}}\|_F^2\right]}{\mathbb{E}\left[\|\mathcal{N}\|_F^2\right]} = \frac{p}{n_b} \tag{5}$$

which is often on the order of tens.

Instead of denoising the original HSI, we denoise the eigenimages in (4). Compared with the use of regularization of self-similar HSI in [16] and [11], the optimization in (4) uses a regularization of self-similar eigenimages, which brings some benefits as follows:

1) High correlation among spectral channels is exploited by minimizing low-rankness or sparsity of a matrix/tensor composed of spectral vectors. The low-rank matrix can be constructed either using pixels from similar patches or using all pixels in the image. The difference between two matrices can be seen in Fig. 2 using the Pavia HSI as an example. The matrix composed of global pixels has far more columns than the matrix constructed using similar patches, implying that the signal subspace in the matrix of global pixels is supported by more observations (i.e., more columns in the matrix) and more observations enable us to have a better estimate of the signal subspace. Therefore, instead of using low-rankness/sparsity of partial pixels (i.e., pixels in nonlocal similar patches) in [16] and [11], we take advantage of the low-rankness of global pixels in our method (4).
2) Eigenimages (of size $r \times c \times p$) have a much smaller amount of data than the original HSI (of size $r \times c \times n_b$). A smaller size of eigenimages leads to a reduction of the computational complexity of eigenimage denoising, which works in a nonlocal patch-based framework.

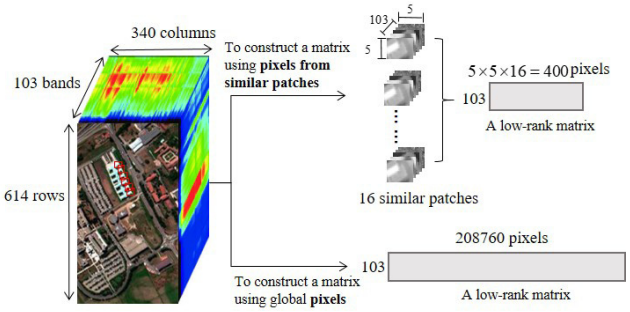


Fig. 2. Two ways to construct low-rank matrices in the Pavia University data set.

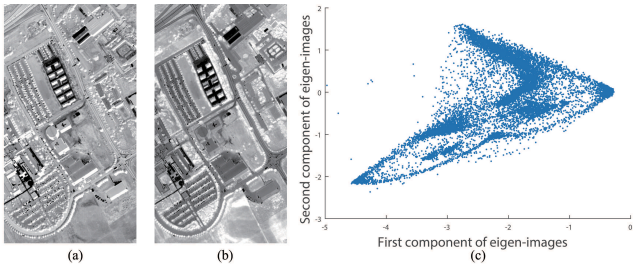


Fig. 3. Pavia HSI. (a) First eigenimage (image of the first representation coefficient); (b) second eigenimage; (c) scattergram of (a)–(b).

The solution of the optimization (4) is

$$\widehat{\mathcal{Z}} = \Psi_{\lambda\phi}\left(\mathcal{Y}\times_3\mathbf{E}^{\mathrm{T}}\right) \tag{6}$$

where $\Psi_{\lambda\phi}(\mathcal{U}) = \arg\min_{\mathcal{X}}(1/2)\|\mathcal{U} - \mathcal{X}\|_F^2 + \lambda\phi(\mathcal{X})$ is the so-called denoising operator, or PO of ϕ [34]. As $\mathbf{E}$ is an orthogonal basis, mode-3 slices of the projected data $\mathcal{Y}\times_3\mathbf{E}^{\mathrm{T}}$ are uncorrelated to each other. Therefore, FastHyDe [5] denoises each mode-3 slice of $\mathcal{Y}\times_3\mathbf{E}^{\mathrm{T}}$ individually.

However, uncorrelation does not imply independence, as illustrated in Fig. 3, where parts (a) and (b) show two eigenimages of the Pavia HSI and part (c) shows the scattergram of the 2-D vectors formed by the two gray levels of the same pixel in (a) and (b). The statistical dependence is clearly observed both in the couple (a)–(b) and in (c). There is still room to improve the HSI denoising performance by taking the statistical dependence among the eigenimages into consideration.

Accordingly, a new method that considers the dependence among eigenimages is proposed for solving (4). The observed eigenimages $\mathcal{Y}\times_3\mathbf{E}^{\mathrm{T}}$ are decomposed into 3-D patches, which are denoised and recombined, yielding estimates $\widehat{\mathcal{Z}}$ and $\widehat{\mathcal{X}} = \widehat{\mathcal{Z}}\times_3\mathbf{E}$. The global structure of the proposed eigenimage denoiser is partially inspired by BM3D [9] and BM4D [11]. The differences concern the formulation with respect to the eigenimages and the filtering, which is adaptive and based on tensor factorization. The details of the proposed method are given in Section III.

III. GLF—The Proposed Method

We propose an HSI denoising algorithm with four steps:

1) subspace identification;
2) grouping similar 3-D patches of eigenimages;

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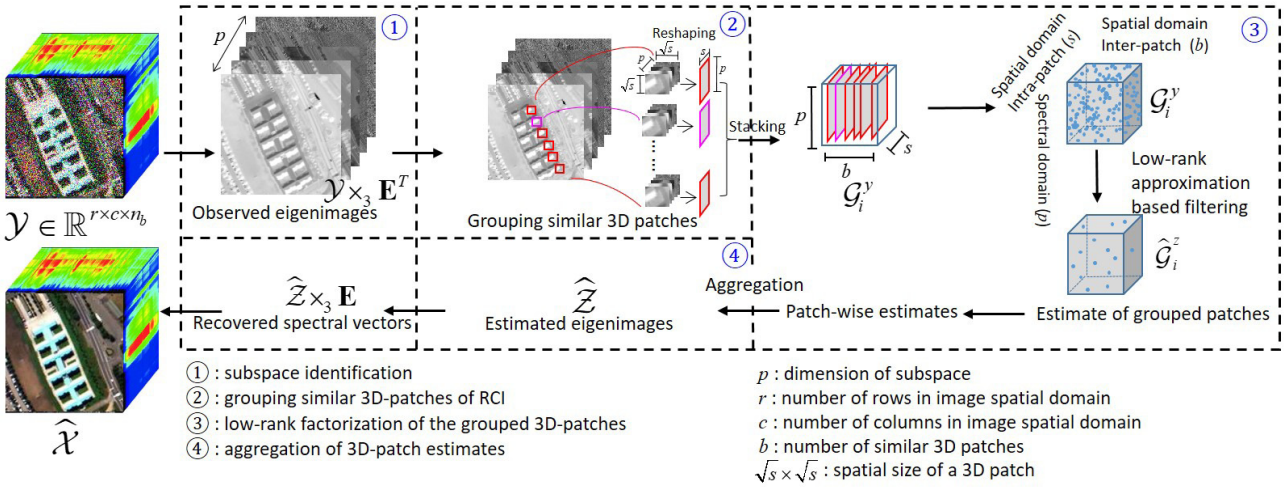


Fig. 4. Flowchart of the proposed denoising algorithm GLF.

3) low-rank factorization of the grouped 3-D patches;
4) aggregation of 3-D patch estimates.

These four steps are shown in Fig. 4. As stated before, the general structure of steps 2, 3, and 4 is inspired by BM4D. The differences reside in step 1, which is not implemented in BM4D, and in the filtering step, which, in our case, is based on the thresholding/shrinkage of SVD singular values and thus data adaptive, whereas in BM4D, it is fixed and based on 3-D discrete cosine transform (DCT) and 1-D Haar wavelets. Below, we detail each step. Given that steps 1 and 3 correspond to a global matrix factorization and local tensor factorizations, respectively, we term our method *global local factorization* (GLF).

A. Step 1: Subspace Identification

The subspace identification step is carried out for i.i.d. noise via the SVD of the spectral correlation matrix $\mathcal{Y}_{(3)}\mathcal{Y}_{(3)}^{\mathrm{T}}/n$. The signal subspace is approximately spanned by p eigenvectors of the spectral correlation matrix corresponding to the top p singular values

$$\mathbf{E} = \mathbf{U}(:, 1:p) \tag{7}$$

where $\mathbf{U} \in \mathbb{R}^{n_b\times n_b}$ is an orthogonal matrix and $\{\mathbf{U}, \Sigma, \mathbf{V}\} = \mathrm{SVD}(\mathcal{Y}_{(3)}\mathcal{Y}_{(3)}^{\mathrm{T}}/n)$ with singular values ordered by nonincreasing magnitude.

For the non-i.i.d. noise, the observation noise is first whitened (see Section III-E). Below, we focus on GLF steps 2, 3, and 4.

B. Step 2: Grouping Similar 3-D Patches of Eigenimages

Let $\mathcal{Y}_i' \in \mathbb{R}^{k\times\sqrt{s}\times\sqrt{s}}$ denote 3-D patches of the projected observed HSI $\mathcal{Y}\times_3\mathbf{E}^{\mathrm{T}}$, where $i \in \{1, \ldots, n\}$ is the index of the top-left pixel, k is the subspace dimension and $\sqrt{s}$ is the spatial size of the 3-D patch (see Fig. 4). Based on the observation model (2), we can write

$$\mathcal{Y}_i' = \mathcal{Z}_i + \mathcal{N}_i' \tag{8}$$

where $\mathcal{Z}_i$ and $\mathcal{N}_i'$ are the corresponding 3-D patches of the clean eigenimages and 3-D patches of projected noise, $\mathcal{N}\times_3\mathbf{E}^{\mathrm{T}}$, respectively.

Given $\mathcal{Y}_i'$, we find, in a neighborhood of pixel i of size 79×79, a set of similar $(b-1)$ 3-D patches, with indexes $i_1, i_2, \ldots, i_{b-1}$, according to the Frobenius distance. With $\mathcal{Y}_i'$ and the $(b-1)$ similar patches, we build a 3-D tensor $\mathcal{G}_i^y \in \mathbb{R}^{k\times b\times s}$ (see step 3 in Fig. 4), such that $\mathcal{G}_i^y(:, j, :) = (\mathcal{Y}_j')_{(1)} \in \mathbb{R}^{k\times s}$, for $j \in \mathcal{I}_i = (i, i_1, i_2, \ldots, i_{b-1})$, and $(\mathcal{Y}_j')_{(1)}$ is the mode-1 matrix unfolding of tensor $\mathcal{Y}_j'$. By construction

$$\mathcal{G}_i^y = \mathcal{G}_i^z + \mathcal{G}_i^n \tag{9}$$

where $\mathcal{G}_i^z$ and $\mathcal{G}_i^n$ are the tensors corresponding to the clean eigenimages and projected noisy 3-D patches, respectively.

C. Step 3: Low-Rank Factorization of Grouped 3-D Patches

The signal component, $\mathcal{G}_i^z$, in (9) is expected to have low rank due to the way that $\mathcal{G}_i^y$, for $i \in \{1, \ldots, n\}$, is constructed. We give an example of the nonlocal low-rankness in Fig. 1, where we randomly selected a 3-D patch of size $5\times 5\times 10$ from the eigenimages of Pavia University data. In the end, 15 patches similar to the target patch were found. A signal tensor $\mathcal{G}^z$ of size $10\times 16\times 25$ was constructed. The low-rankness of the tensor $\mathcal{G}^z$ can be inferred from the plots of singular values of unfolded mode-1, mode-2, and mode-3 matrices. The first (the largest) singular value dominates all other singular values, which decrease to near-zero rapidly.

However, the noise component, $\mathcal{G}_i^n$, in (9) is expected to have higher rank, which might be as high as $\min\{ks, kb, sb\}$ [35]. On the other hand, the lower the rank of the approximation is, the higher the noise attenuation. Therefore, by finding the lowest rank approximations for the 3-D tensors $\mathcal{G}_i^y$, we maximally attenuate the noise while preserving the signal. A low-rank approximation $\widehat{\mathcal{G}}_i^z$ of $\mathcal{G}_i^z$ from $\mathcal{G}_i^y$, with the ability to minimize the mean square error $\|\widehat{\mathcal{G}}_i^z - \mathcal{G}_i^z\|_F$, is computed and is a low-rank tensor approximation problem [35].

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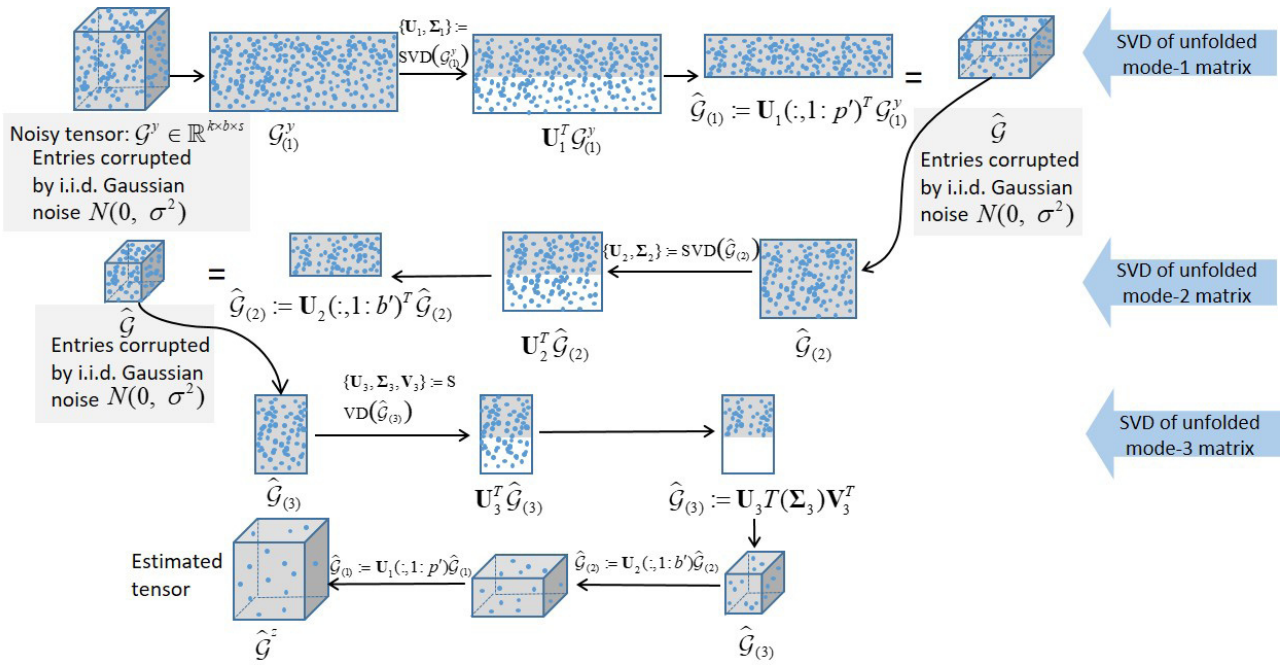


Fig. 5. Flowchart of the low-rank factorization of grouped 3-D patches.

The low-rank tensor approximation problem is a more complicated and challenging problem compared to the low-rank matrix approximation. We prefer to convert the low-rank tensor approximation problem to a low-rank matrix approximation problem, because in the case of matrices and using results from random matrix theory [36], the low-rank approximations for matrices with additive i.i.d. Gaussian noise may be optimally computed in an asymptotic sense. This computation is carried out via SVD by properly thresholding and shrinking its singular values [36], [37]. In Appendix B, we provide details of the two LRMR methods used in this work.

We exploit the results in Appendix B by successively applying them to the mode-1, mode-2, and mode-3 matrix unfoldings of tensor $\mathcal{G}^y \in \mathbb{R}^{k\times b\times s}$, as shown in Algorithm 1 and Fig. 5. The spatial indexes in the subscript of $\mathcal{G}_i^y$ are omitted for simplicity.

Line 3 of Algorithm 1 computes the projection coefficients of the columns of $\mathcal{G}^y_{(1)}$ onto the range of $\mathbf{U}_1(:, 1 : p')$, where $(\sigma_1, \ldots, \sigma_p) = \text{diag}(\Sigma_1)$ and $p' = \arg\min_l t(\sigma_l)$ are subject to $t(\sigma_l) > 0$. The rank of the unfolding matrix $\mathcal{G}^y_{(1)}$ can be inferred from the Shrinkage function $t(\cdot)$ of the singular values, which is based on random matrix theory [36], [37], and given in Appendix B.

Line 6 of Algorithm 1 is similar to line 3 applied to dimension 2 of the tensor $\widehat{\mathcal{G}}$. In Line 9, we apply not only thresholding as before but also the shrinkage as defined in expression (19) in Appendix B [37]. In lines 11 and 12, the tensor is projected back to the original dimensions.

We remark that the distribution of the noise variable does not change when we perform subspace projection and then successively implement the SVD of the unfolded mode-1, mode-2, and mode-3 matrix unfoldings of tensor $\mathcal{G}^y$ (see Fig. 5). Assume that the noise in the observed image is i.i.d. and follows a Gaussian distribution with zero mean and variance σ^2, i.e., $n_{ijk} \sim N(0, \sigma^2)$. We successively perform orthogonal transforms of the observed image in different dimensions. Since the transform matrices $\mathbf{E}$, $\mathbf{U}_1$, $\mathbf{U}_2$, and $\mathbf{U}_3$ are orthogonal, the noise in the transformed data is still i.i.d. and follows a Gaussian distribution. This is important because it enables the subsequent application of the same filtering thresholding results based on random matrix theory [36], [37].

Algorithm 1 Successive SVD Low-Rank Factorization[1]

1: Filtering in the spectral domain
2: $\{\mathbf{U}_1, \Sigma_1\} := \text{SVD}(\mathcal{G}^y_{(1)})$, $\mathcal{G}^y_{(1)} \in \mathbb{R}^{p\times(bs)}$
3: $\widehat{\mathcal{G}}_{(1)} := \mathbf{U}_1(:, 1 : p')^{\text{T}}\mathcal{G}^y_{(1)}$, $\widehat{\mathcal{G}} \in \mathbb{R}^{p'\times b\times s}$, $p' \le p$
4: Filtering in the spatial domain (interpatch)
5: $\{\mathbf{U}_2, \Sigma_2\} := \text{SVD}(\widehat{\mathcal{G}}_{(2)})$, $\widehat{\mathcal{G}}_{(2)} \in \mathbb{R}^{b\times(p's)}$
6: $\widehat{\mathcal{G}}_{(2)} := \mathbf{U}_2(:, 1 : b')^{\text{T}}\widehat{\mathcal{G}}_{(2)}$, $\widehat{\mathcal{G}} \in \mathbb{R}^{p'\times b'\times s}$, $b' \le b$
7: Filtering in the spatial domain (intrapatch)
8: $\{\mathbf{U}_3, \Sigma_3, \mathbf{V}_3\} := \text{SVD}(\widehat{\mathcal{G}}_{(3)})$, $\widehat{\mathcal{G}}_{(3)} \in \mathbb{R}^{s\times(p'b')}$
9: $\widehat{\mathcal{G}}_{(3)} := \mathbf{U}_3\ T(\Sigma_3)\mathbf{V}_3^{\text{T}}$, $\widehat{\mathcal{G}} \in \mathbb{R}^{p'\times b'\times s}$
10: Inverse transform to compute $\widehat{\mathcal{G}}^z$
11: $\widehat{\mathcal{G}}_{(2)} := \mathbf{U}_2(:, 1 : b')\widehat{\mathcal{G}}_{(2)}$, $\widehat{\mathcal{G}} \in \mathbb{R}^{p'\times b\times s}$
12: $\widehat{\mathcal{G}}_{(1)} := \mathbf{U}_1(:, 1 : p')\widehat{\mathcal{G}}_{(1)}$, $\widehat{\mathcal{G}} \in \mathbb{R}^{p\times b\times s}$
13: **return** $\widehat{\mathcal{G}}^z := \widehat{\mathcal{G}}$

D. Step 4: Aggregation of 3-D Patch Estimates

The estimated 3-D patches contained in tensors $\widehat{\mathcal{G}}_i^z$ for $i = 1, \ldots, n$ are added back to their original location in the eigenimages and then averaged taking into account that a given pixel belongs to various 3-D patches. Finally, we compute $\widehat{\mathcal{X}} = \widehat{\mathcal{Z}} \times_3 \mathbf{E}$.

[0]A MATLAB demo of GLF is available at https://github.com/LinaZhuang/HSI-denoiser-GLF.git

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E. Additive Gaussian Non-I.I.D. Noise

In real HSIs, however, noise tends to be non-i.i.d.; that is, it is pixelwise independent but bandwise dependent. We assume that the spectral covariance $\mathbf{C}_\lambda$ of the noise in the real HSIs is positive definite and therefore nonsingular. We can convert the non-i.i.d. scenario into an i.i.d. scenario by using data whitening

$$\widetilde{\mathcal{Y}} := \mathcal{Y} \times_3 \sqrt{\mathbf{C}_\lambda^{-1}} \tag{10}$$

where $(\mathbf{C}_\lambda)^{1/2}$ and $\mathbf{C}_\lambda^{-1}$ denote the square root of $\mathbf{C}_\lambda$ and the inverse of $\mathbf{C}_\lambda$, respectively.

As a preprocessing step, (10) converts the noise so that it becomes standard Gaussian and i.i.d. The image conversion (10) can be rewritten in matrix form

$$\widetilde{\mathbf{Y}} = \sqrt{\mathbf{C}_\lambda^{-1}}\mathbf{X} + \sqrt{\mathbf{C}_\lambda^{-1}}\mathbf{N}. \tag{11}$$

The spectral covariance of the noise in the converted image is

$$\begin{aligned}\widetilde{\mathbf{C}}_\lambda &= \mathbb{E}\left[\sqrt{\mathbf{C}_\lambda^{-1}}\mathbf{n}_i\left(\sqrt{\mathbf{C}_\lambda^{-1}}\mathbf{n}_i\right)^{\mathrm{T}}\right] = \sqrt{\mathbf{C}_\lambda^{-1}}\mathbb{E}\left[\mathbf{n}_i\mathbf{n}_i^{\mathrm{T}}\right]\sqrt{\mathbf{C}_\lambda^{-1}}^{\mathrm{T}} \\ &= \sqrt{\mathbf{C}_\lambda^{-1}}\mathbf{C}_\lambda\sqrt{\mathbf{C}_\lambda^{-1}}^{\mathrm{T}} = \mathbf{I}.\end{aligned} \tag{12}$$

The noise in the whitened image is standard Gaussian and i.i.d. Therefore, the converted image $\widetilde{\mathcal{Y}}$ can be denoised as discussed in Section IIIA–D. Finally, a clean image is recovered as

$$\widehat{\mathcal{X}} = \widehat{\widetilde{\mathcal{X}}} \times_3 \sqrt{\mathbf{C}_\lambda} \tag{13}$$

where $\widehat{\widetilde{\mathcal{X}}}$ is the estimated clean version of image $\widetilde{\mathcal{Y}}$.

IV. Experiments With Simulated Images

Both simulated and real data experiments are carried out to assess the performance of the proposed GLF for HSI noise removal. Eight state-of-the-art HSI denoising methods are selected for comparison. The results are evaluated with respect to both the visualization and numerical measures.

A. Simulation of Noisy Images and Comparisons

1) Simulation of Noisy Images: Two public hyperspectral data sets were employed to simulate the noisy images:

1) The first data set is the Washington DC Mall data set, which was collected by the hyperspectral digital imagery collection experiment, and contains $1209 \times 307 \times 191$ entries. A subregion image of size $200 \times 200 \times 191$ is used in our experiment. Spectral vectors (i.e., mode-3 vectors) of the subregion image were projected onto the 8-D signal subspace learned by performing an SVD on the spectral correlation matrix. The obtained HSI is considered to be a clean HSI.
2) The generation of the second simulated image is proposed in [38] using the spatial structure of an image from the real Indian Pines data set. Pixels with the same label are assigned with a specific spectral signature extracted from the United States Geological Survey (USGS) digital spectral library. The synthetic HSI of size $145 \times 145 \times 224$ is considered a clean HSI. This simulated Indian Pines image[2] has been used in many works to test HSI denoisers, for example, double-factor regularized low-rank tensor factorization (LRTF-DFR) [2], total variation regularized low-rank tensor decomposition (LRTDTV) [39], and weighted group sparsity-regularized low-rank tensor decomposition (LRTDGS) [40].
3) Finally, the values of two clean HSIs are normalized into [0, 1], and Gaussian i.i.d. noise with standard deviation $\sigma = \{0.02; 0.04; 0.06; 0.08; 0.10\}$ was added to simulate noisy HSIs.

2) Comparisons: To thoroughly evaluate the performance of the proposed method, eight state-of-the-art HSI denoising methods are selected for comparison, i.e., BM4D [11], LRMR [18], NAIRLMA [19], total variation-regularized low-rank matrix factorization (LRTV) [38], LRTF-DFR [2], FastHyDe [5],[3] "Subspace+BM4D" (which is BM4D applied to subspace representation coefficients) [11], and WLRTR [16]. The methods BM4D, LRMR, NAIRLMA, and LRTV are well-known benchmarks for the comparison of hyperspectral denoisers. FastHyDe and "Subspace+BM4D" are subspace-based filtering methods similar to GLF. LRTF-DFR and WLRTR are newly proposed methods based on low-rank tensor factorization. All experiments are implemented in MATLAB (R2019a) on Windows 10 with an Intel Core i7-8700 3.20-GHz processor and 16-GB RAM.

For quantitative assessment, the peak signal-to-noise ratio (PSNR) index and the structural similarity (SSIM) index [10] of each band are calculated. The mean PSNR (MPSNR) and mean SSIM (MSSIM) results of the compared methods on the two data sets are presented in Table I to evaluate the efficiency of the proposed method. The computation time (in seconds) was also calculated to evaluate the performance of the methods. In particular, we highlight the best results in Table I in bold.

B. Denoising Performance

First, we discuss the parameters used in the proposed method and the compared methods. The noise variance used in the methods BM4D, FastHyDe, "Subspace+BM4D," and GLF are estimated by HySime [24]. The subspace dimension parameter used in methods FastHyDe, "Subspace+BM4D," and GLF methods was set according to the noise standard deviation σ. For the Washington DC Mall data, the dimension of the subspace was set to $\{8; 7; 7; 6; 5\}$ for images with noise standard deviation $\sigma \in \{0.02; 0.04; 0.06; 0.08; 0.10\}$. For the simulated Indian Pines data, the dimension of the subspace was set to $\{17; 15; 15; 15; 15\}$ for the simulated images with $\sigma \in \{0.02; 0.04; 0.06; 0.08; 0.10\}$. The nonlocal patch-based denoisers, including BM4D, FastHyDe, and "Subspace+BM4D," have some parameters that are related to the grouping of similar patches, for example, patch-size ($\sqrt{s} \times \sqrt{s}$), size of the search region ($N_S \times N_S$), the sliding step used to process the next reference patch (N_{step}), and

[2]The simulated clean Indian Pines data is available at https://github.com/YuBangZheng/code_LRTFDFR

[3]The MATLAB codes of FastHyDe denoising method are available at https://github.com/LinaZhuang/FastHyDe_FastHyIn

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TABLE I
QUANTITATIVE ASSESSMENT OF DIFFERENT DENOISERS APPLIED TO TWO SIMULATED IMAGES IN TERMS OF MPSRN (IN dB), MSSIM, AND TIME

noise std σ	Indexes	Noisy	BM4D [11]	LRMR [18]	NAIRLMA [19]	LRTV [38]	LRTF-DFR [2]	FastHyDe [5]	Subspace +BM4D [11]	WLRTR [16]	GLF
Simulated Washington DC Mall data											
0.02	MPSNR	33.97	44.70	46.55	50.79	45.72	44.44	54.39	54.36	51.50	**55.09**
	MSSIM	0.6430	0.8807	0.9457	0.9692	0.9510	0.9636	0.9941	0.9939	0.9799	**0.9950**
	Times(s)	-	188	110	91	49	45	5	**4**	1066	178
0.04	MPSNR	27.95	40.73	42.28	45.57	42.50	44.15	49.47	49.31	46.87	**50.15**
	MSSIM	0.5414	0.8131	0.8879	0.9368	0.9082	0.9616	0.9856	0.9842	0.9557	**0.9875**
	Times(s)	-	187	105	58	53	45	**4**	**4**	1535	172
0.06	MPSNR	24.43	38.52	39.80	42.82	40.67	43.79	46.29	46.04	44.15	**46.82**
	MSSIM	0.4694	0.7670	0.8432	0.9125	0.8756	0.9583	0.9755	0.9716	0.9304	**0.9773**
	Times(s)	-	187	103	47	51	49	4	**3**	1528	169
0.08	MPSNR	21.93	36.99	37.99	40.96	39.30	43.38	44.59	44.04	42.37	**45.07**
	MSSIM	0.4110	0.7334	0.8096	0.8933	0.8486	0.9541	0.9659	0.9570	0.9157	**0.9688**
	Times(s)	-	187	103	37	51	48	4	**3**	2259	165
0.10	MPSNR	19.99	35.81	36.55	39.33	38.32	42.76	43.18	42.40	40.86	**43.71**
	MSSIM	0.3621	0.7062	0.7824	0.8697	0.8291	0.9480	0.9578	0.9435	0.8940	**0.9603**
	Times(s)	-	188	102	36	51	49	3	**2**	2280	163
Simulated Indian Pines data											
0.02	MPSNR	33.98	50.22	46.83	49.74	51.34	47.11	54.90	56.54	58.65	**59.49**
	MSSIM	0.8318	0.9981	0.9959	0.9938	0.9971	0.9983	0.9994	0.9997	**0.9999**	0.9998
	Times(s)	-	115	58	64	41	28	**4**	7	636	70
0.04	MPSNR	27.95	45.12	42.98	44.20	46.18	46.94	49.36	50.76	51.61	**53.12**
	MSSIM	0.6440	0.9937	0.9871	0.9796	0.9912	0.9982	0.9979	0.9989	**0.9992**	**0.9992**
	Times(s)	-	114	61	43	38	28	**4**	6	901	70
0.06	MPSNR	24.43	42.18	39.98	41.46	42.77	46.65	46.19	47.38	47.67	**49.42**
	MSSIM	0.5244	0.9877	0.9737	0.9655	0.9816	0.9981	0.9959	0.9976	0.9977	**0.9981**
	Times(s)	-	115	59	33	48	27	**4**	6	898	71
0.08	MPSNR	21.93	40.08	37.75	39.32	40.34	46.23	43.93	44.96	45.16	**46.69**
	MSSIM	0.4434	0.9808	0.9568	0.9483	0.9689	**0.9979**	0.9937	0.9961	0.8229	0.9966
	Times(s)	-	114	59	30	42	28	**4**	5	1346	70
0.10	MPSNR	20.00	38.44	36.01	37.70	38.58	**45.78**	42.30	43.23	42.95	44.85
	MSSIM	0.3836	0.9732	0.9371	0.9310	0.9551	**0.9977**	0.9914	0.9946	0.9928	0.9949
	Times(s)	-	114	60	27	40	32	**3**	5	1565	70

the (maximum) cardinality of similar patch groups (N_2). Parameters N_S, N_{step}, and N_2 were fixed to 78, 3, and 16, respectively, in all experiments. The patch-size value was set to five for the simulated Indian Pines data and ten for the Washington DC Mall data. The parameters of the other compared methods were set according to the suggestion in the corresponding literature.

The denoising performance in terms of MPSNR, MSSIM, and computation time (in seconds) is presented in Table I, in which we highlight the best results in bold. The MPSNR and MSSIM results of the proposed GLF are higher than those of the other methods in most cases. Basically, the nonlocal patch-based methods, including FastHyDe, "Subspace+BM4D," WLRTR, and GLF, outperform the other methods, namely, LRMR, NAILRMA, LRTV, and LRTF-DFR. The reason for this is that the nonlocal patch-based framework can take full advantage of the high spatial correlation in natural images by grouping similar patches. The grouping of similar patches greatly enhances the sparsity of the data to be filtered, implying that a larger amount of noise can be removed. However, the only exception is BM4D, which does not perform as well as the other nonlocal patch-based methods in the experiments. We remark that BM4D is designed for general high-dimensional data, such as video data and magnetic resonance images. The size setting of similar cubes in BM4D only use partial bands, not all bands. Therefore, BM4D works well in general high-dimensional data but is not optimal for HSI denoising. We consider BM4D as the benchmark in our experiments. Among the nonlocal patch-based methods, the proposed GLF shares a similar processing framework to "Subspace+BM4D." Both of them work on the coefficients of the subspace, where they search for similar patches. The key difference is that the "Subspace+BM4D" sparsely represents the groups of similar patches on a fixed basis, whereas the GLF sparsely represents the groups on a data-adaptive basis (learned by SVD). The results in Table I show that the data-adaptive basis is a very compact and efficient representation [41], that leads to the better denoising performance of GLF.

The PSNR and SSIM values of each restoration band of simulated Washington DC Mall images and simulated Indian Pines images are depicted in Figs. 6 and 7, respectively. We can see that the performance of BM4D, LRMR, NAIRLMA, LRTV, LRTF-DFR, and WLRTR varies greatly at different noise levels and data. Clearly, our proposed GLF, shown in red, uniformly provides the best performances in most cases.

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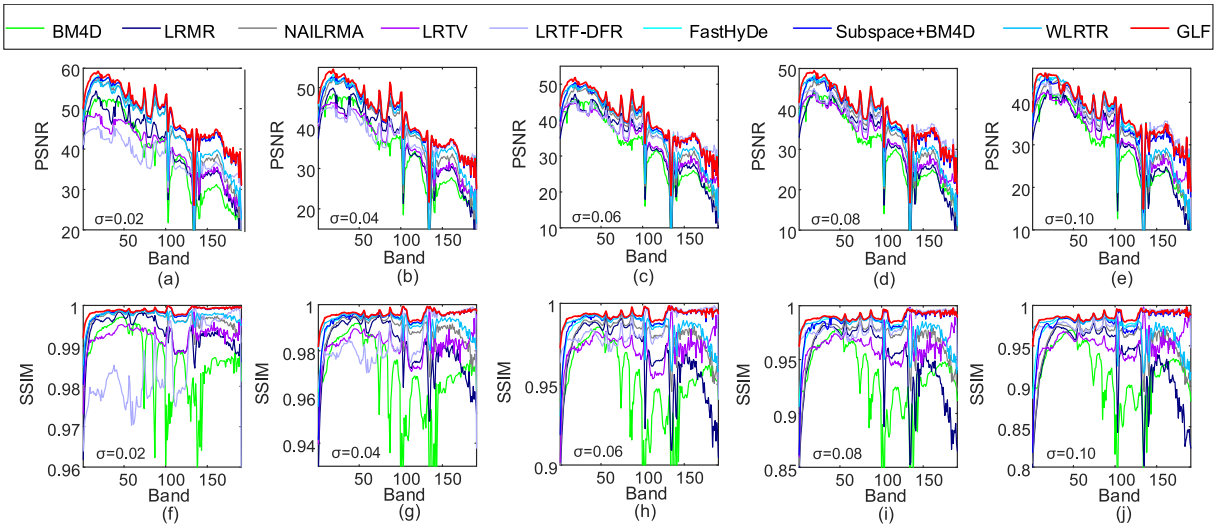


Fig. 6. PSNR and SSIM values of each band of denoised Washington DC Mall images in the cases with different noise levels: (a)–(e) are PSNR in the cases with noise σ=0.02, noise σ=0.04, noise σ=0.06, noise σ=0.08, and noise σ=0.10, respectively; (f)–(j) are SSIM in the cases with noise σ=0.02, noise σ=0.04, noise σ=0.06, noise σ=0.08, and noise σ=0.10, respectively.

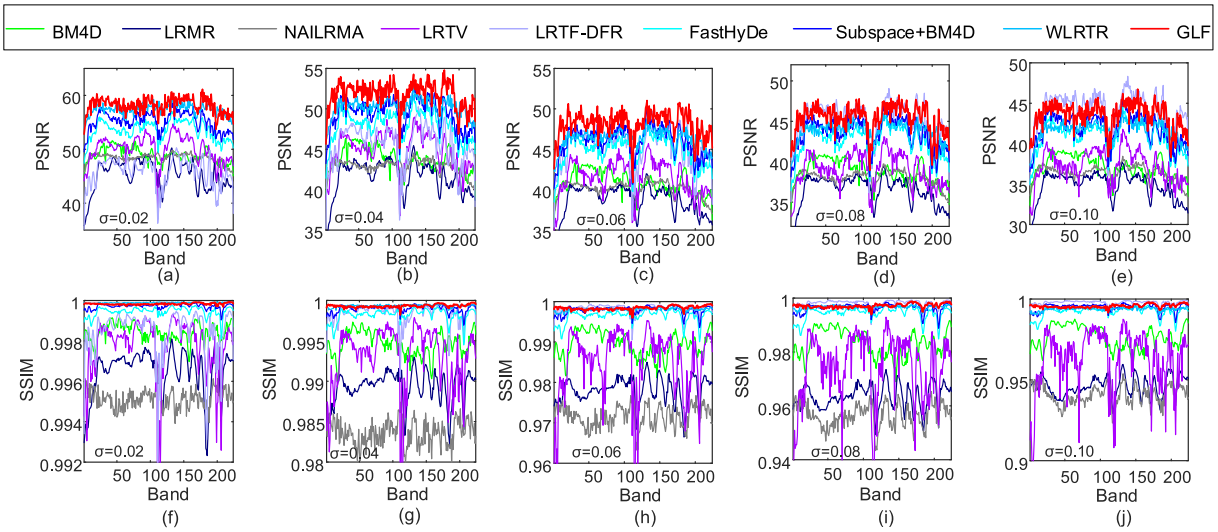


Fig. 7. PSNR and SSIM values of each band of restoration of simulated Indian Pines images in the cases with different noise levels: (a)–(e) are PSNR in the cases with noise σ=0.02, noise σ=0.04, noise σ=0.06, noise σ=0.08, and noise σ=0.10, respectively; (f)–(j) are SSIM in the cases with noise σ=0.02, noise σ=0.04, noise σ=0.06, noise σ=0.08, and noise σ=0.10, respectively.

For a visual comparison, Figs. 8 and 9 display band 3 of denoised Washington DC Mall images in case $\sigma = 0.06$ and band 106 of simulated Indian Pines images in case $\sigma = 0.06$. To highlight the differences among the denoising results, we focus on small regions containing detailed structures, which are circled in red. We can see from Fig. 8 that BM4D, LRTV, LRTF-DFR obtain over-smoothed results, as some textures circled in red are lost. If we focus on the small region containing homogeneous materials in the first circle, then the results of LRMR, NAILRMA, FastHyDe, and "Subspace+BM4D" still retain a small amount of noise. WLRTR and GLF yield the best results in Fig. 8. Fig. 9 also zooms in on a small region, where we can see that the results of WLRTR and GLF are visually closer to the clean one than other methods.

To compare the denoising results more clearly, we generate a difference image by computing $\text{abs}(\widehat{\mathcal{X}}(:,:,i) - \mathcal{X}(:,:,i))$, $i = 1, \ldots, n_b$, which is the absolute value of the difference between the estimated image and its corresponding clean image. The smaller the difference is, the better the denoising performance. Difference images of the Washington DC Mall image are shown in Fig. 10, where we can see that the images of LRTF-DFR, WLRTR, and GLF are darker than the images of other methods in general. Furthermore, the difference image of GLF has less structure than those of LRTF-DFR and WLRTR, whose difference images still contain some details of the image. Difference images of the simulated Indian Pines data are shown in Fig. 11, where we can see that the difference images of LRMR, NAILRMA, and LRTV still contain random Gaussian noise. The difference images of the remaining denoisers display large estimation errors on the edges of landcovers; however, the estimation errors of the proposed GLF are smaller than those of the other methods.

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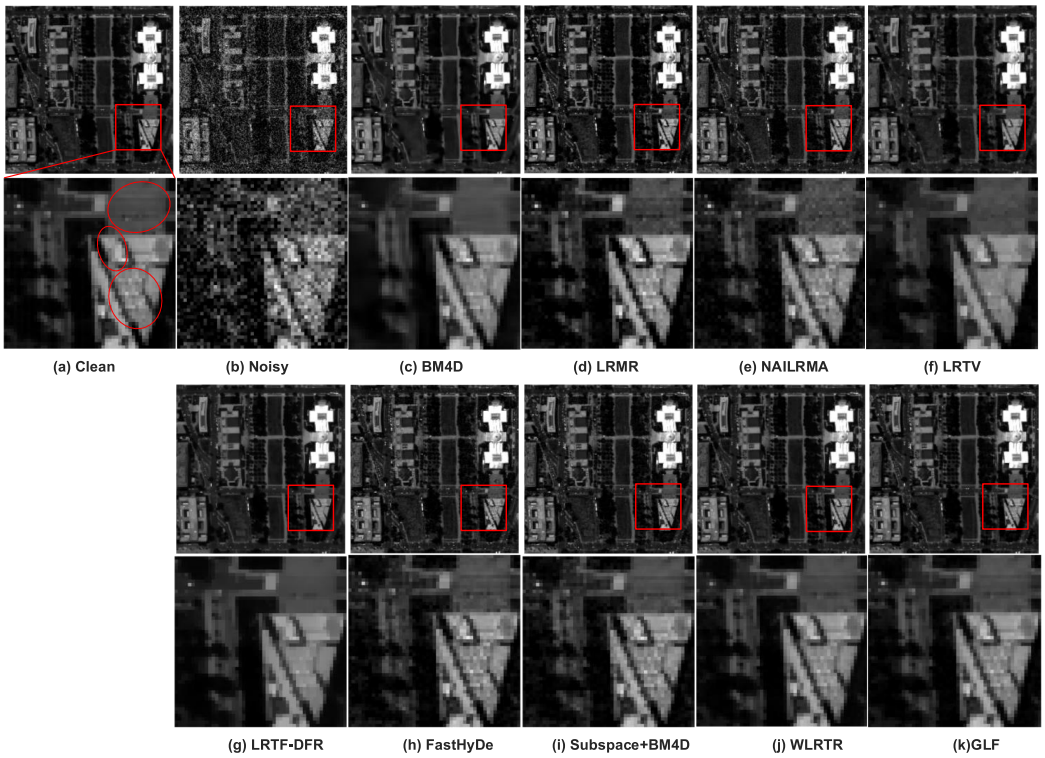


Fig. 8. Denoising band 3 of the Washington DC Mall image in the case $\sigma = 0.06$: (a) Clean image; (b) Noisy image; (c) BM4D; (d) LRMR; (e) NAIRLMA; (f) LRTV; (g) LRTF-DFR; (h) FastHyDe; (i) Subspace+BM4D; (j) WLRTR; (k) GLF.

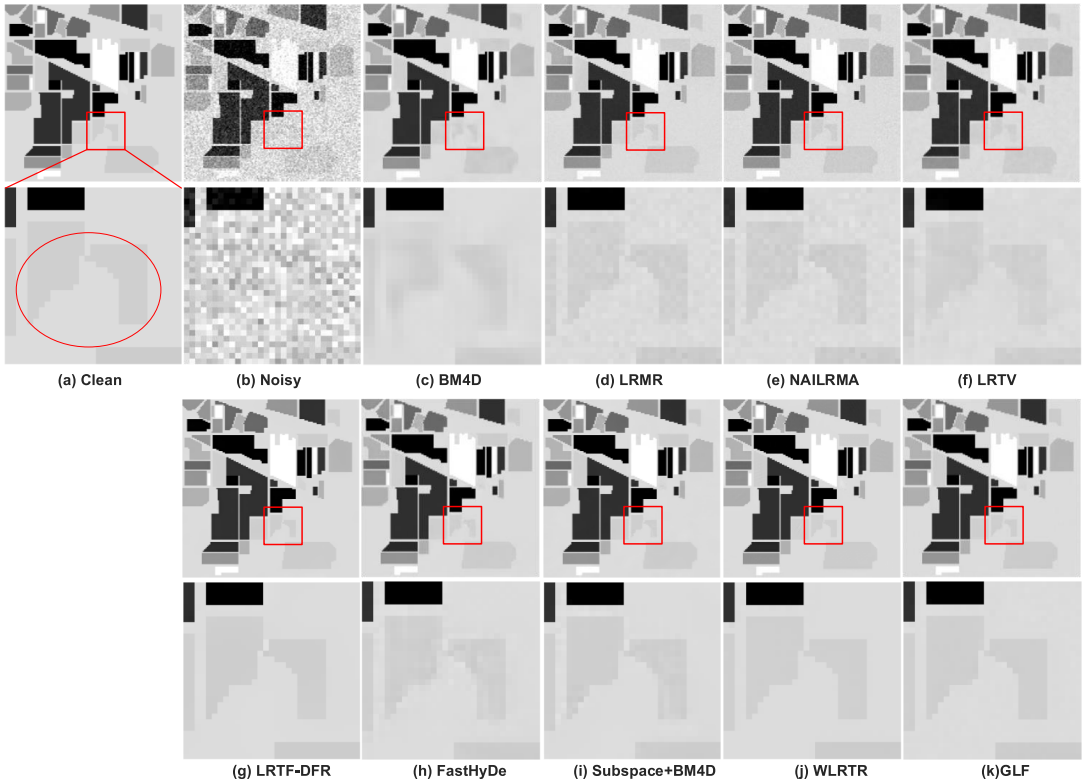


Fig. 9. Denoising band 106 of the Indian Pines image in the case $\sigma = 0.06$; (a) Clean image; (b) Noisy image; (c) BM4D; (d) LRMR; (e) NAIRLMA; (f) LRTV; (g) LRTF-DFR; (h) FastHyDe; (i) Subspace+BM4D; (j) WLRTR; (k) GLF.

As the spectral information of HSIs is of critical importance for the identification of materials, we compare the denoising results of spectral vectors. Figs. 12 and 13 show denoised spectral vectors of one pixel in the Washington DC Mall image and one in the simulated Indian Pines image, respectively. It is observed that the spectral vectors recovered by the proposed

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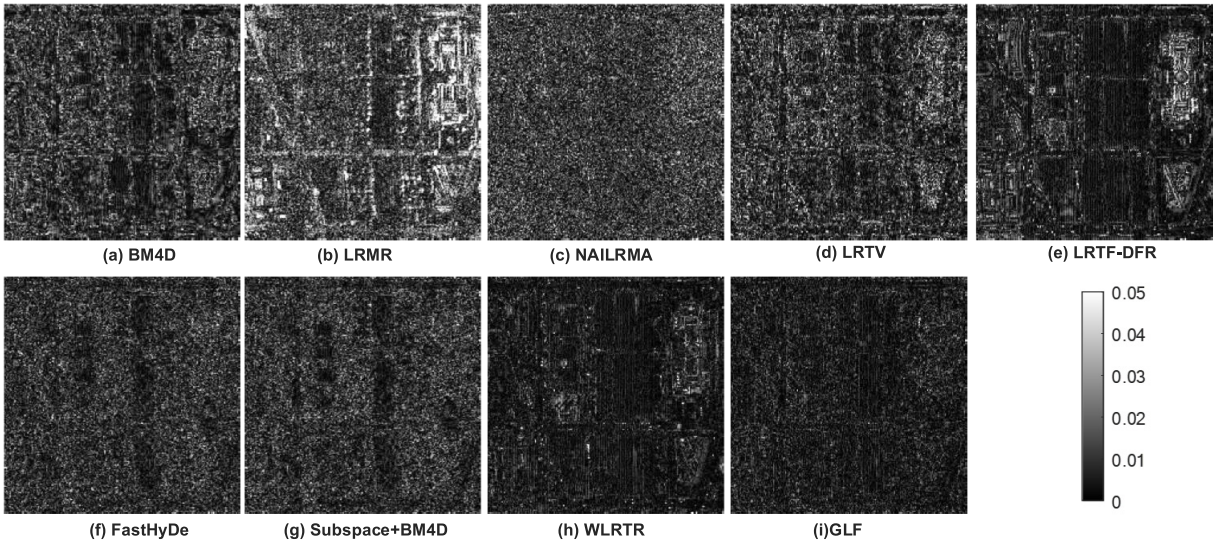


Fig. 10. Difference images generated by computing abs($\widehat{\mathcal{X}}(:,:,3) - \mathcal{X}(:,:,3)$) on the simulated Washington DC Mall data: (a) BM4D; (b) LRMR; (c) NAIRLMA; (d) LRTV; (e) LRTF-DFR; (f) FastHyDe; (g) Subspace+BM4D; (h) WLRTR; (i) GLF.

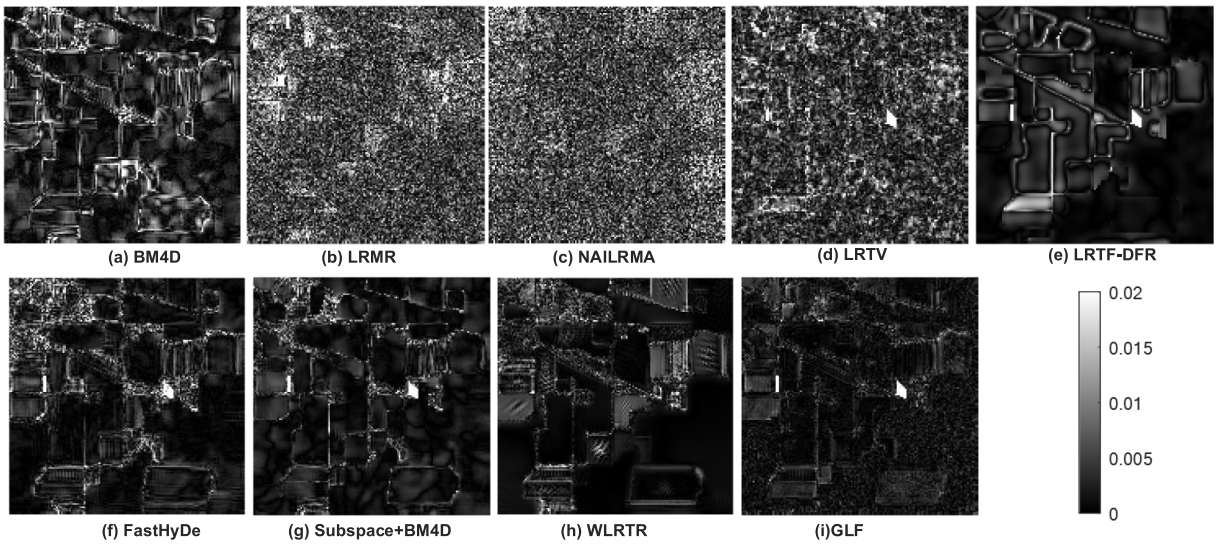


Fig. 11. Difference images generated by computing abs($\widehat{\mathcal{X}}(:,:,106) - \mathcal{X}(:,:,106)$) on the simulated Indian Pines data: (a) BM4D; (b) LRMR; (c) NAIRLMA; (d) LRTV; (e) LRTF-DFR; (f) FastHyDe; (g) Subspace+BM4D; (h) WLRTR; (i) GLF.

GLF can better approximate the clean vectors than those produced by the other methods.

To summarize, the abovementioned observations suggest that the proposed GLF performs better than the other methods and is capable of better preserving the image structure information and spectral signatures. The reason why our proposed method is superior to the compared methods is that it takes full advantage of the global low-rank structure of HSIs in the spectral domain and local low-rank structure of nonlocal similar image patches.

C. Comparison Between Eigenimage Denoising and HSI Denoising

By taking advantage of the global low-rankness of HSIs in the spectral domain, the proposed GLF performs nonlocal patch-based denoising on eigenimages instead of the original HSI. If we set the transform matrix $\mathbf{E} = \mathbf{I}_{n_b}$ in the proposed GLF method, then we actually perform nonlocal patch-based denoising on the original HSI. Fig. 14 provides results of the GLF method with the setting $\mathbf{E} = \mathbf{I}_{n_b}$, compared with the proposed GLF method. We can see the use of eigenimages greatly improves the denoising performance in terms of MPSNR and reduces the computational complexity during processing.

D. Parameter Analysis

As a nonlocal patch-based framework, the proposed GLF method contains some parameters related to how to search for similar patches. Among the patch-related parameters, the size of the search region ($N_S \times N_S$), sliding step to process every next reference patch (N_{step}), and (maximum) cardinality of similar patch groups (N_2) mainly affect the running time rather than denoising accuracy. For example, Fig. 15 shows the denoising performance of the proposed GLF with different settings of N_2 in two simulated images. We can see the setting of N_2 does not significantly affect denoising results

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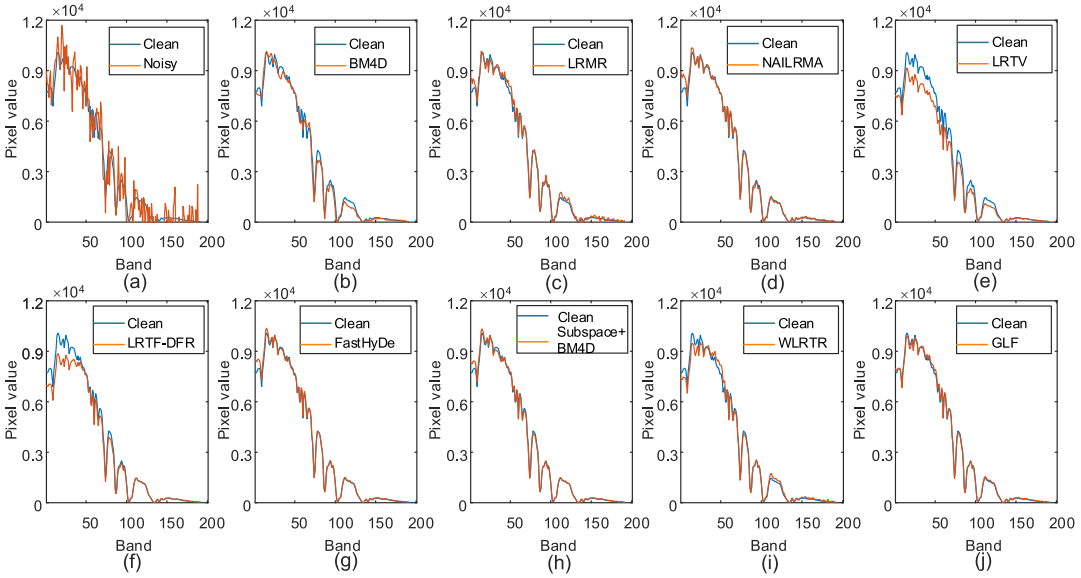


Fig. 12. Spectral vector of one pixel and its denoising results in the simulated Washington DC Mall image with $\sigma = 0.06$. Spectral signatures were plotted using the original values, instead of normalized values: (a) Noisy; (b) BM4D; (c) LRMR; (d) NAILRMA; (e) LRTV; (f) LRTF-DFR; (g) FastHyDe; (h) Subspace+BM4D; (i) WLRTR; (j) GLF.

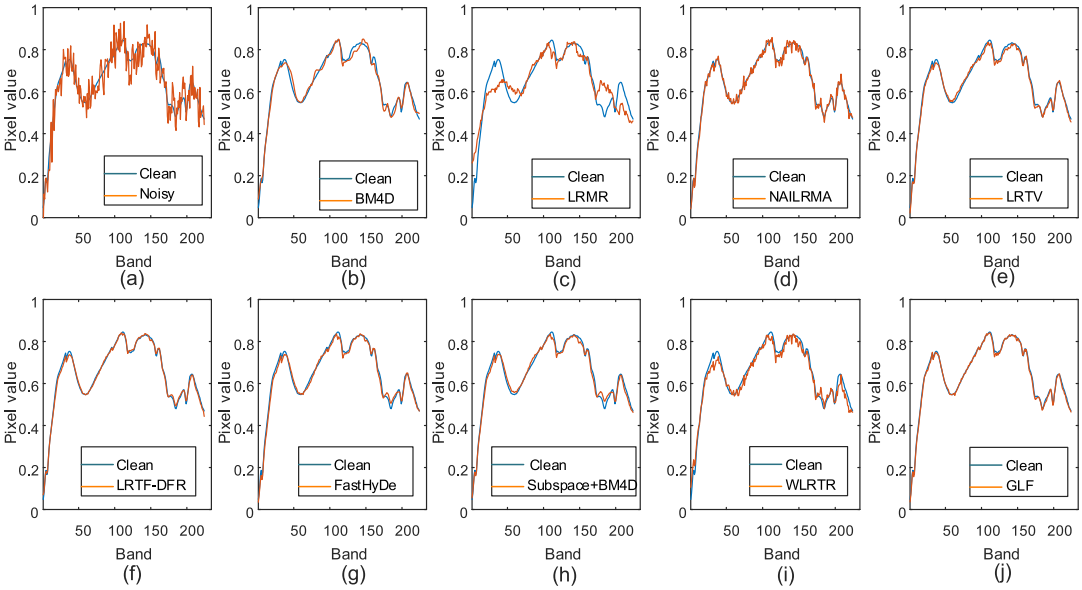


Fig. 13. Spectral vector of one pixel and its denoising results in the simulated Indian Pines image with $\sigma = 0.06$: (a) Noisy; (b) BM4D; (c) LRMR; (d) NAILRMA; (e) LRTV; (f) LRTF-DFR; (g) FastHyDe; (h) Subspace+BM4D; (i) WLRTR; (j) GLF.

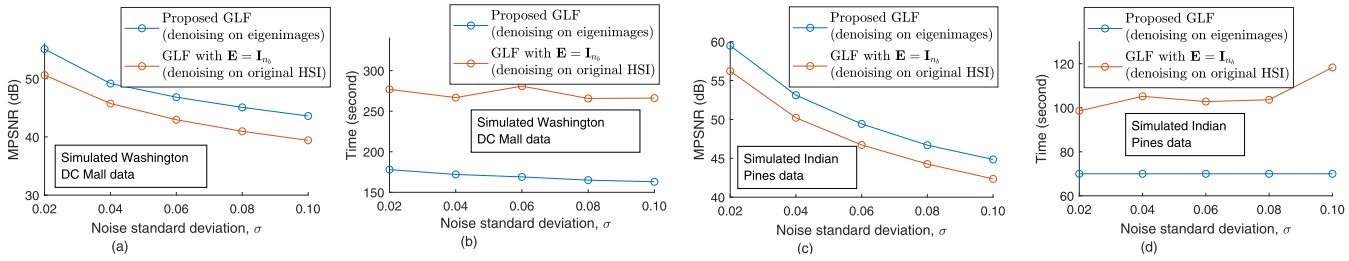


Fig. 14. A performance comparison of the proposed GLF method and the GLF with $\mathbf{E} = \mathbf{I}_{n_b}$ (i.e., denoising on the original HSI) in terms of MPSNR in (a), (c) and running time in (b), (d).

in terms of MPSNR. Thus, we simply set these parameters to the same values for all nonlocal patch-based denoisers in our experiments.

Apart from the parameters mentioned above, the proposed GLF has critical parameters, namely, the variance of Gaussian noise, σ^2, size of patch, $\sqrt{s} \times \sqrt{s}$, and dimension

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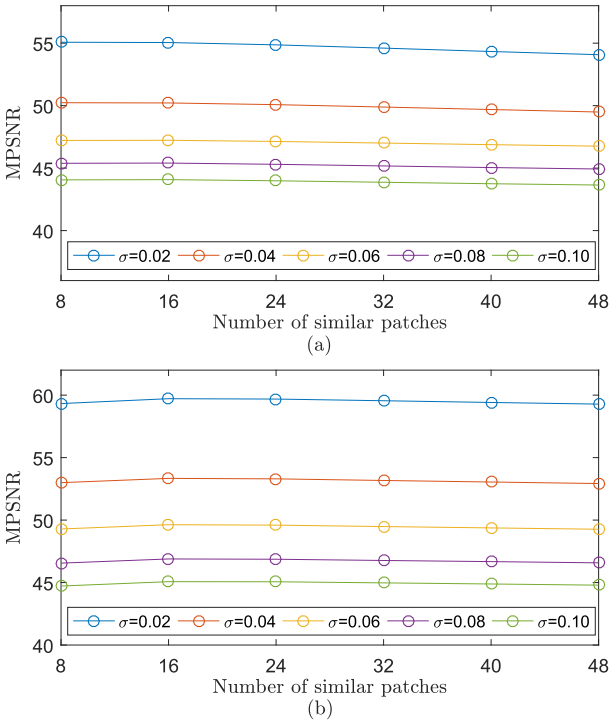


Fig. 15. MPSNR values of noisy Washington DC Mall images with various noise variances σ^2 in (a) and of the simulated Indian Pines images in (b) denoised by GLF as a function of the parameter, (maximum) cardinality of similar patches.

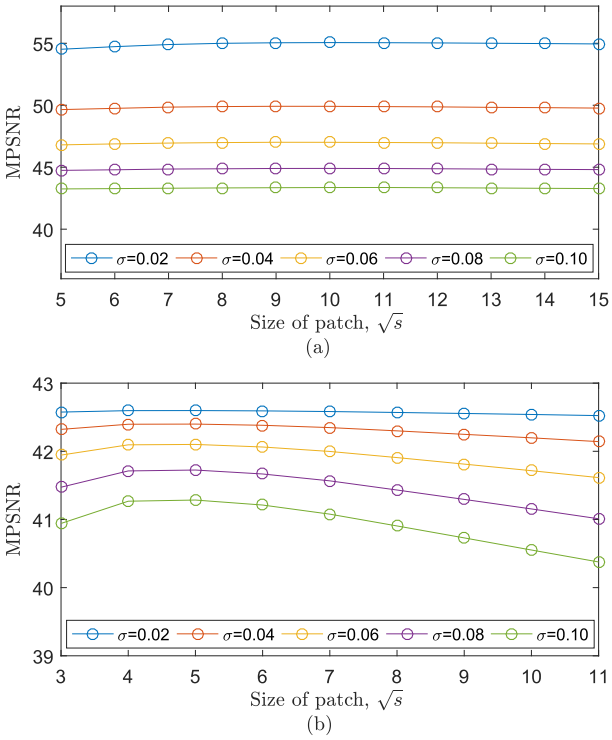


Fig. 16. MPSNR values (in dB) of noisy Washington DC Mall images with various noise variances σ^2 in (a) and of the simulated Indian Pines images in (b) denoised by GLF as a function of the parameter, patch-size.

of subspace, p. We estimate the noise variance of HSIs by HySime [24].

The optimal size of the patch, $\sqrt{s}$, should be set adaptively according to the images. Specifically, the optimal value of

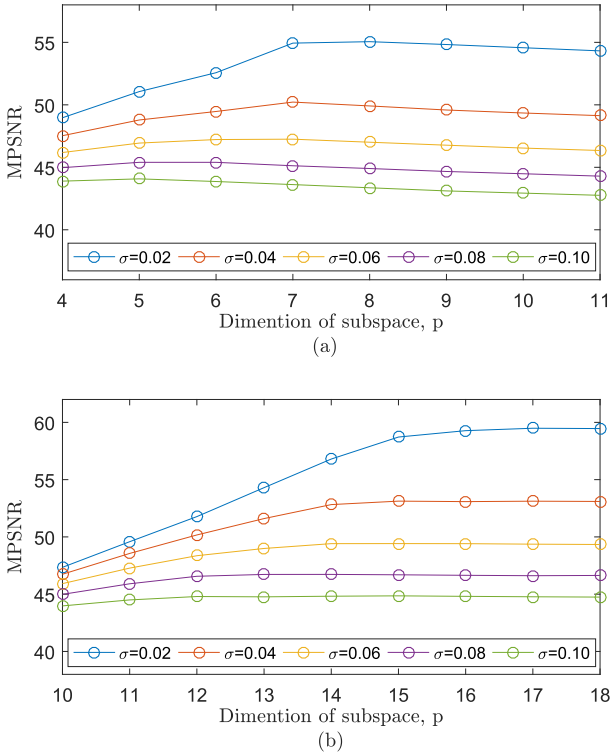


Fig. 17. MPSNR values of noisy Washington DC Mall images with various noise variances σ^2 in (a) and of the simulated Indian Pines images in (b) denoised by GLF as a function of the parameter, dimension of subspace.

$\sqrt{s}$ usually depends on the spatial resolution of the images, landcover types, and variance of the noise. We analyze the impact of $\sqrt{s}$ on GLF denoising in Fig. 16, where the MPSNR values in the first subfigure are stable while the size of the patch varies from 5×5 to 15×15 in the simulated Washington DC Mall image. However, Fig. 16(b) shows that the optimal size of the patch is approximately 5×5 when the noise variances in the simulated Indian Pines images are relatively high. Therefore, we suggest that the setting of patch-size should be adaptive according to the noise variance in the experiments.

The impact of the dimension of the subspace, p, on GLF denoising in terms of the MPSNR is shown in Fig. 17. We can see that the MPSNR is practically constant provided that the dimension of the subspace is not underestimated. This means that GLF is relatively robust to the overestimation of the subspace dimension.

V. Experiments With Real Images

A. AVIRIS Indian Pines Data

The real HSI in the Indian Pines data[4] was recorded over northwestern Indiana in June 1992. The image has 145×145 pixels in the spatial domain and 220 channels in the spectral domain, with a spatial resolution of 20 m per pixel. A number of bands of this image are corrupted by noise.

[4]The real Indian Pines image is available at https://engineering.purdue.edu/biehl/MultiSpec/hyperspectral.html

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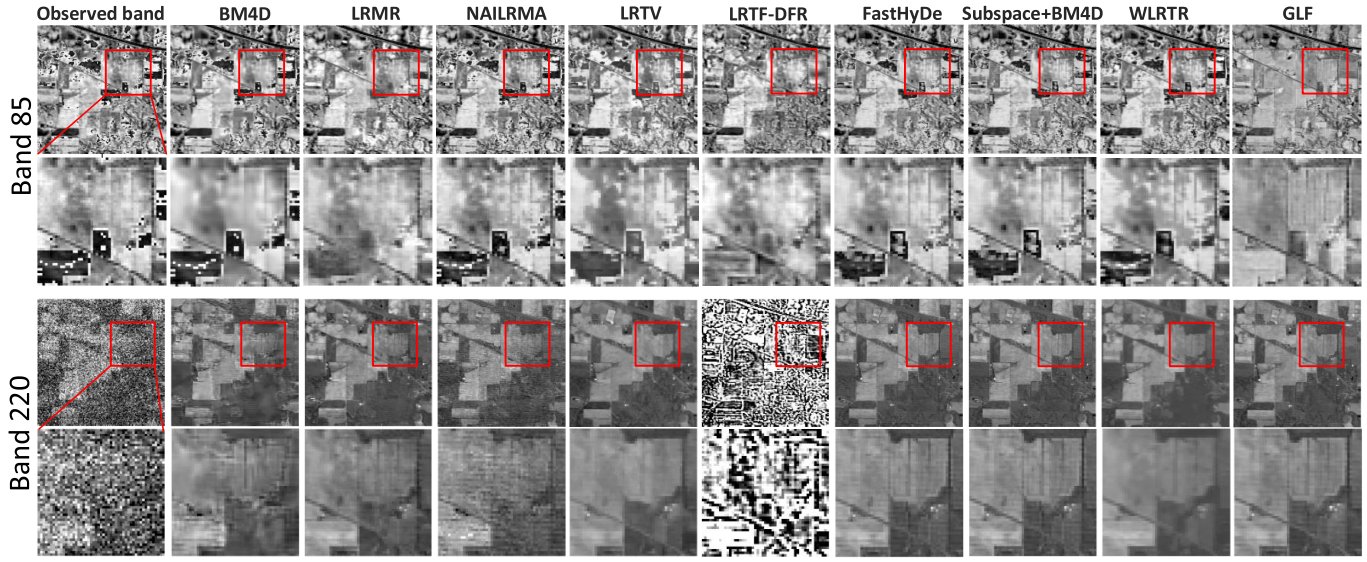


Fig. 18. Denoised bands 85 and 220 of the real Indian Pines image.

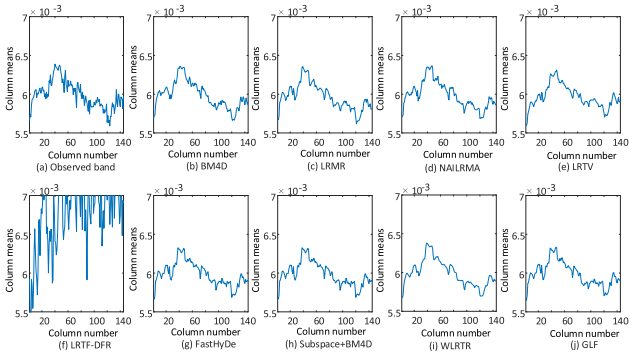


Fig. 19. Column mean curves of observed band 220 and its denoising results in the real Pines image: (a) Observed band; (b) BM4D; (c) LRMR; (d) NAIRLMA; (e) LRTV; (f) LRTF-DFR; (g) FastHyDe; (h) Subspace+BM4D; (i) WLRTR; (j) GLF.

We display the denoising results of two typical bands, namely, band 220 and band 85 in Fig. 18. Due to the lack of a clean counterpart to this image, we have to compare the denoising results visually. Band 85 has severe noise, but its noise may be eliminated by exploiting the correlation among bands. The proposed GLF yields the best result for band 85 visually. The column mean curves of band 220 are presented in Fig. 19. In band 220, the denoised results of BM4D, LRMR, NAILRMA, LRTV, and LRTF-DFR are still noisy, while the results of and WLRTR are oversmooth. The results of FastHyDe, "Subspace+BM4D," and GLF in band 220 are visually better than other methods. The proposed GLF manages to recover local details, especially edges.

B. SpecTIR Data

The second real HSI was acquired by a data collection campaign called the SpecTIR hyperspectral airborne Rochester experiment (SHARE) on 29 July 2010, with the ProSpecTIR-VS2 sensor. A subset of the data with a size of $180 \times 180 \times 126$ was used as the testing image. The image displays strong noise in a number of bands.

We show three noisy bands, namely, bands 50, 83, and 120, and their denoised results in Fig. 20, where we can see that the Gaussian noise and slight stripes result in the corruption of the SpecTIR data set. It is clear that BM4D, LRMR, NAILRMA, LRTV, LRTF-DFR, and WLRTR are not able to remove the noise thoroughly. These results are consistent with the column mean curves of band 120 shown in Fig. 21. FastHyDe and "Subspace+BM4D" filter the image excessively, leading to blurring results. The proposed GLF makes a good compromise between filtering and preserving the detailed structure.

VI. Conclusion

We proposed a new HSI denoiser, GLF, which represents HSIs in low-dimensional subspaces. The denoising is formulated in the subspace representation coefficients. Considering that components of the subspace representation coefficients are self-similar and statistically dependent, they are jointly denoised in a nonlocal patch-based framework. Furthermore, a new filter based on low-rank approximation is devised for filtering noisy similar 3-D patches. In comparison with the state-of-the-art HSI denoisers, GLF uniformly outperformed the competitors.

Appendix A
Proof of the Equivalence of (3) and (4)

Let $\mathbf{E}^{\perp} \in \mathbb{R}^{n_b \times (n_b - p)}$ be the orthogonal complement of the subspace spanned by the columns of matrix $\mathbf{E}$; then, we have $\mathbf{E}\mathbf{E}^{\mathrm{T}} + \mathbf{E}^{\perp}\mathbf{E}^{\perp T} = \mathbf{I}_{n_b}$, and $\mathcal{Y} = \mathcal{Y} \times_3 \mathbf{I}_{n_b} = \mathcal{Y} \times_3 \mathbf{E}\mathbf{E}^{\mathrm{T}} + \mathcal{Y} \times_3 \mathbf{E}^{\perp}\mathbf{E}^{\perp T}$.

We can equivalently reformulate (3) as the following optimization:

$$\widehat{\mathcal{Z}} = \arg\min_{\mathcal{Z}} \frac{1}{2}\|\mathcal{Y} - \mathcal{Z} \times_3 \mathbf{E}\|_F^2 + \lambda\phi(\mathcal{Z}) \tag{3}$$

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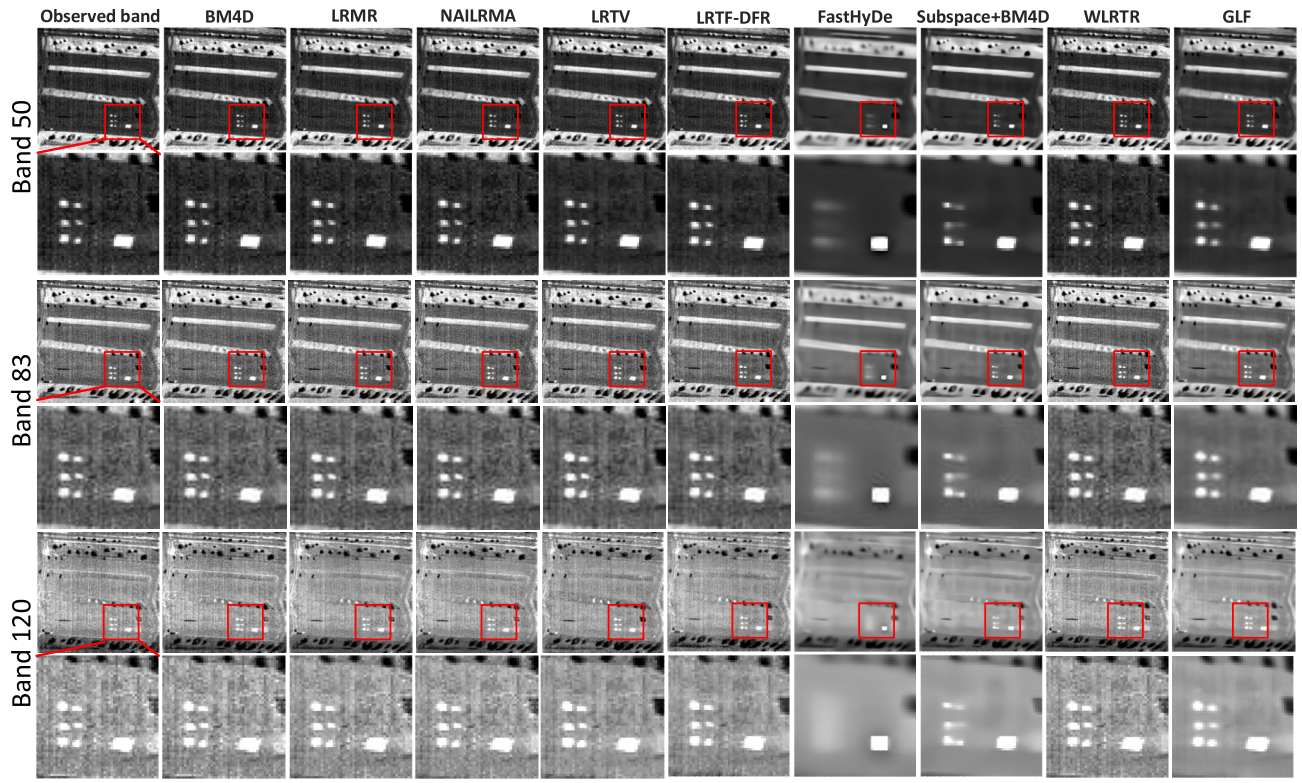


Fig. 20. Denoised bands 50, 83, and 120 of the real SpecTIR image.

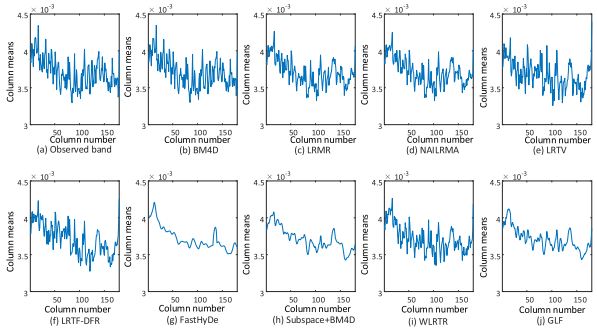


Fig. 21. Column mean curves of observed band 120 and its denoising results in the real SpecTIR image: (a) Observed band; (b) BM4D; (c) LRMR; (d) NAIRLMA; (e) LRTV; (f) LRTF-DFR; (g) FastHyDe; (h) Subspace+BM4D; (i) WLRTR; (j) GLF.

$$= \arg\min_{\mathcal{Z}} \frac{1}{2}\|\mathcal{Y}\times_3 \mathbf{E}\mathbf{E}^{\mathrm{T}} + \mathcal{Y}\times_3 \mathbf{E}^{\perp}\mathbf{E}^{\perp T} - \mathcal{Z}\times_3 \mathbf{E}\|_F^2 + \lambda\phi(\mathcal{Z}) \tag{3a}$$

$$= \arg\min_{\mathcal{Z}} \frac{1}{2}\|\mathcal{Y}\times_3 \mathbf{E}\mathbf{E}^{\mathrm{T}} - \mathcal{Z}\times_3 \mathbf{E}\|_F^2 + \lambda\phi(\mathcal{Z}) \tag{3b}$$

$$= \arg\min_{\mathcal{Z}} \frac{1}{2}\|\left(\mathcal{Y}\times_3 \mathbf{E}^{\mathrm{T}} - \mathcal{Z}\right)\times_3 \mathbf{E}\|_F^2 + \lambda\phi(\mathcal{Z}) \tag{3c}$$

$$= \arg\min_{\mathcal{Z}} \frac{1}{2}\|\mathcal{Y}\times_3 \mathbf{E}^{\mathrm{T}} - \mathcal{Z}\|_F^2 + \lambda\phi(\mathcal{Z}). \tag{4}$$

Optimized (3a) is equivalent to (3b) because $\mathcal{Y}\times_3 \mathbf{E}^{\perp}\mathbf{E}^{\perp T}$, representing the signal component of observed spectral vectors lying on the orthogonal complement of the subspace $\mathbf{E}$, is a constant when solving the optimization with respect to $\mathbf{Z}$.

Optimization (3c) is equivalent to (4) due to the isometry property. As mentioned previously, columns of the matrix $\mathbf{E} \in \mathbb{R}^{n_b \times p}(n_b > p)$ are orthogonal, i.e., $\mathbf{E}^{\mathrm{T}}\mathbf{E} = \mathbf{I}_p$, which implies the isometry property

$$\|\mathbf{E}\mathbf{x}\|_2 = \|\mathbf{x}\|_2, \text{ for all } \mathbf{x} \text{ in } \mathbb{R}^p. \tag{14}$$

The proof is completed.

Appendix B
Low-Rank Matrix Recovery by Optimal Shrinker

Given a noisy observation $\mathbf{Y} \in \mathbb{R}^{r\times c}$ of a latent low-rank matrix $\mathbf{X} \in \mathbb{R}^{r\times c}$, the observation model can be formulated as

$$\mathbf{Y} = \mathbf{X} + \mathbf{N} \tag{15}$$

where $\mathbf{N} \in \mathbb{R}^{r\times c}$ denotes the noise matrix with Gaussian i.i.d. entries $n_{i,j} \sim \mathcal{N}(0,\ 1)$.

LRMR aims to estimate $\mathbf{X}$, given $\mathbf{Y}$. Here, we give two approaches used in this work for LRMR. The main idea is to truncate or shrink the singular values of the observed matrix $\mathbf{Y}$. The SVD of $\mathbf{Y}$ is a factorization of the form

$$\mathbf{Y} = \mathbf{U}\Sigma\mathbf{V}^{\mathrm{T}} \tag{16}$$

where $\mathbf{U} \in \mathbb{R}^{r\times c}$ and $\mathbf{V} \in \mathbb{R}^{c\times c}$ are unitary matrices, and $\Sigma = \mathrm{Diag}[\sigma_1, \sigma_2, \ldots] \in \mathbb{R}^{r\times c}$ is a rectangular matrix with nonnegative real numbers (called singular values) on the diagonal.

A low-rank matrix approximating $\mathbf{Y}$ can be obtained by using the truncated SVD of $\mathbf{Y}$

$$\begin{aligned}\widehat{\mathbf{X}} &= \mathbf{U}(:, 1:k)\Sigma(1:k, 1:k)\mathbf{V}(:, 1:k)^{\mathrm{T}} \\ &= \mathbf{U}(:, 1:k)\mathbf{U}(:, 1:k)^{\mathrm{T}}\mathbf{Y}\end{aligned} \tag{17}$$

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where only k column vectors of $\mathbf{U}$ and k row vectors of $\mathbf{V}^{\mathrm{T}}$ corresponding to the k largest singular values are calculated.

Instead of using a thresholding function on singular values, Shabalin and Nobel proposed a LRMR method using a shrinkage function on singular values [36], [37]. An estimate of $\mathbf{X}$ is given by

$$\widehat{\mathbf{X}} = \mathbf{U}T(\Sigma)\mathbf{V}^{\mathrm{T}} \tag{18}$$

where $T(\Sigma) = \mathrm{Diag}[t(\sigma_{\mathbf{Y},1}), t(\sigma_{\mathbf{Y},2}), \dots]$, with $t : \mathbb{R}_0^+ \to \mathbb{R}_0^+$ is the shrinkage function. We use the subscript in $\sigma_{\mathbf{Y},i}$ to indicate that it is the ith singular value of matrix $\mathbf{Y}$. By analyzing the effects of noise on the SVD of matrix $\mathbf{X}$, the optimal shrinkage function is obtained as

$$t\left(\sigma_{\mathbf{Y},i}\right) = \begin{cases} \hat{\sigma}_{\mathbf{X},i}\hat{\theta}_i\hat{\phi}_i, & \text{if } \sigma_{\mathbf{Y},i} > 1+\sqrt{\beta} \\ 0, & \text{others} \end{cases} \tag{19}$$

where

$$\frac{r}{c} \to \beta \tag{20}$$

$$\hat{\sigma}^2_{\mathbf{X},i} = \frac{1}{2}\left[\left(\sigma^2_{\mathbf{Y},i} - 1 - \beta\right) + \sqrt{\left(\sigma^2_{\mathbf{Y},i} - 1 - \beta\right)^2 - 4\beta}\right] \tag{21}$$

$$\hat{\theta}_i^2 = \left(1 - \frac{\beta}{\hat{\sigma}^4_{\mathbf{X},i}}\right) \Big/ \left(1 + \frac{\beta}{\hat{\sigma}^2_{\mathbf{X},i}}\right) \tag{22}$$

and,

$$\hat{\phi}_i^2 = \left(1 - \frac{\beta}{\hat{\sigma}^4_{\mathbf{X},i}}\right) \Big/ \left(1 + \frac{1}{\hat{\sigma}^2_{\mathbf{X},i}}\right). \tag{23}$$

ACKNOWLEDGMENT

The authors would like to thank the Jet Propulsion Laboratory for generously providing the airborne visible/infrared imaging spectrometer (AVIRIS) data and the United States Geological Survey (USGS) for their publicly available library of mineral signatures.

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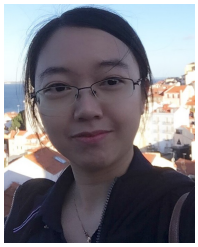
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Lina Zhuang (Member, IEEE) received the bachelor's degrees in geographic information system and in economics from South China Normal University, Ghuangzhou, China, in 2012, the M.S. degree in cartography and geography information system from the Institute of Remote Sensing and Digital Earth, Chinese Academy of Sciences, Beijing, China, in 2015, and the Ph.D. degree in electrical and computer engineering from the Instituto Superior Tecnico, Universidade de Lisboa, Lisbon, Portugal, in 2018.

Since 2015, she has been a Marie Curie Early Stage Researcher of Sparse Representations and Compressed Sensing Training Network (SpaRTaN number 607290) with the Instituto de Telecomunicações, Lisbon. SpaRTaN Initial Training Networks (ITN) is funded under the European Union's Seventh Framework Programme (FP7-PEOPLE-2013-ITN) call and is part of the Marie Curie Actions-ITN funding scheme. She is a Research Assistant Professor with Hong Kong Baptist University, Kowloon, Hong Kong. Her research interests include hyperspectral image restoration, superresolution, and compressive sensing.



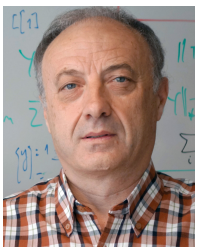
Xiyou Fu received the bachelor's degrees in remote sensing science and technology from Wuhan University, Wuhan, China, in 2012, the M.S. degree in electronic and communication engineering and the Ph.D. degree in cartography and geography information system from Institute of Remote Sensing and Digital Earth, Chinese Academy of Sciences, Beijing, China, in 2015 and 2019, respectively. He is currently a Post-Doctoral Researcher with Shenzhen University, Shenzhen, China. His research interests include hyperspectral image restoration, anomaly detection, and superresolution.



Michael K. Ng (Senior Member, IEEE) received the B.Sc. and M.Phil. degrees from the University of Hong Kong, Pokfulam, Hong Kong, in 1990 and 1992, respectively, and the Ph.D. degree from the Chinese University of Hong Kong, Shatin, Hong Kong, in 1995.

He is a Chair Professor with the Department of Mathematics, University of Hong Kong. His research interests include bioinformatics, image processing, scientific computing, and data mining.

Dr. Ng serves for the editorial boards of international journals.



José M. Bioucas-Dias (Fellow, IEEE) received the E.E., M.Sc., Ph.D., and Habilitation degrees in electrical and computer engineering from Instituto Superior Técnico (IST), Universidade Técnica de Lisbon (now Universidade de Lisboa), Lisbon, Portugal, in 1985, 1991, 1995, and 2007, respectively.

Since 1995, he had been with the Department of Electrical and Computer Engineering, IST, where he was a Professor and teaches inverse problems in imaging and electric communications. He was also a Senior Researcher with the Pattern and Image Analysis Group of the Instituto de Telecomunicações, Lisbon, which is a private nonprofit research institution. His research interests include inverse problems, signal and image processing, pattern recognition, optimization, and remote sensing.

Dr. Bioucas-Dias had introduced scientific contributions in the areas of imaging inverse problems, statistical image processing, optimization, phase estimation, phase unwrapping, and in various imaging applications, such as hyperspectral and radar imaging. He was included in Thomson Reuters' Highly Cited Researchers 2015 and 2018 lists and received the IEEE GRSS David Landgrebe Award for 2017.

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